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**Related U.S. Application Data**

division of application No. 16/284,468, filed on Feb. 25, 2019, now Pat. No. 10,985,206, which is a continuation of application No. PCT/KR2017/009562, filed on Aug. 31, 2017.

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(51) **Int. Cl.**

*H10H 20/831* (2025.01)  
*H10H 29/14* (2025.01)

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FIG. 1

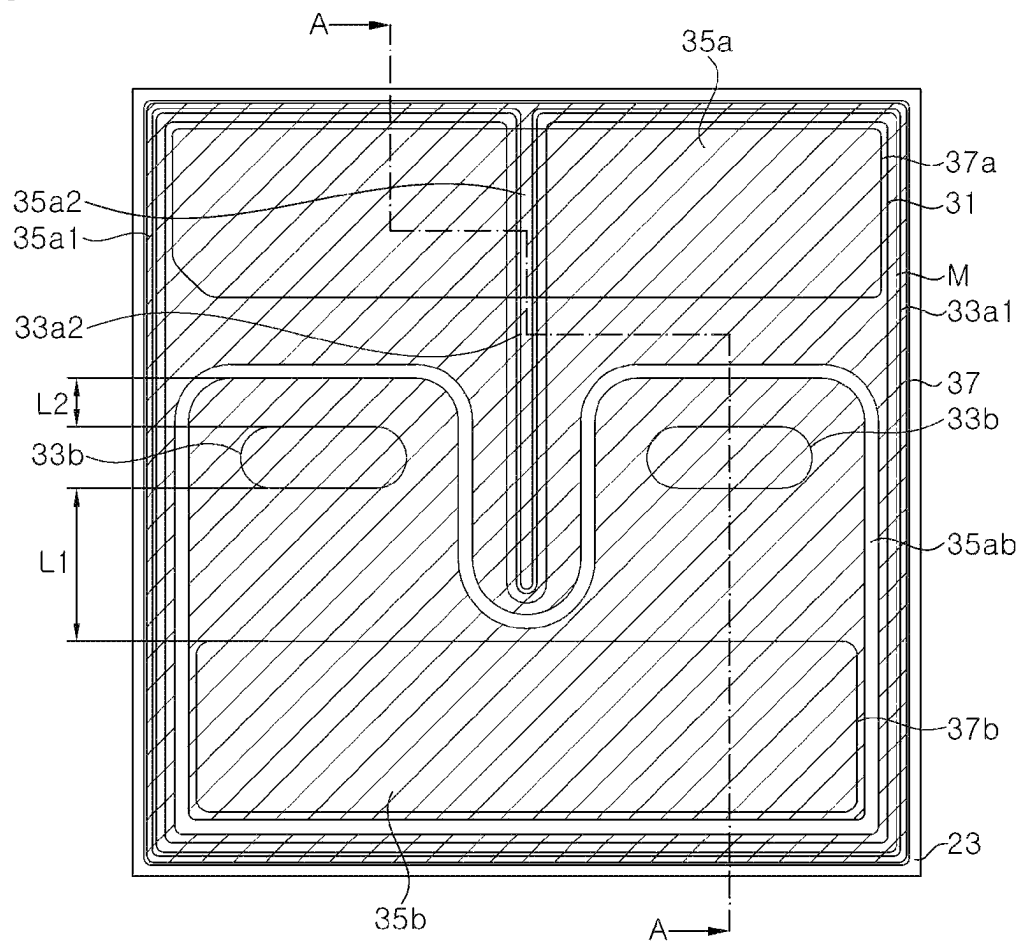


FIG. 2

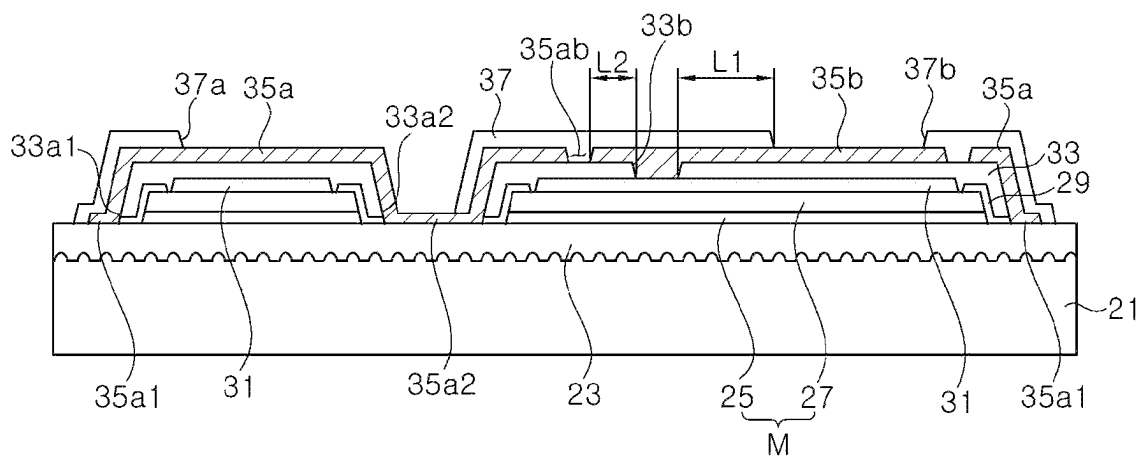


FIG. 3

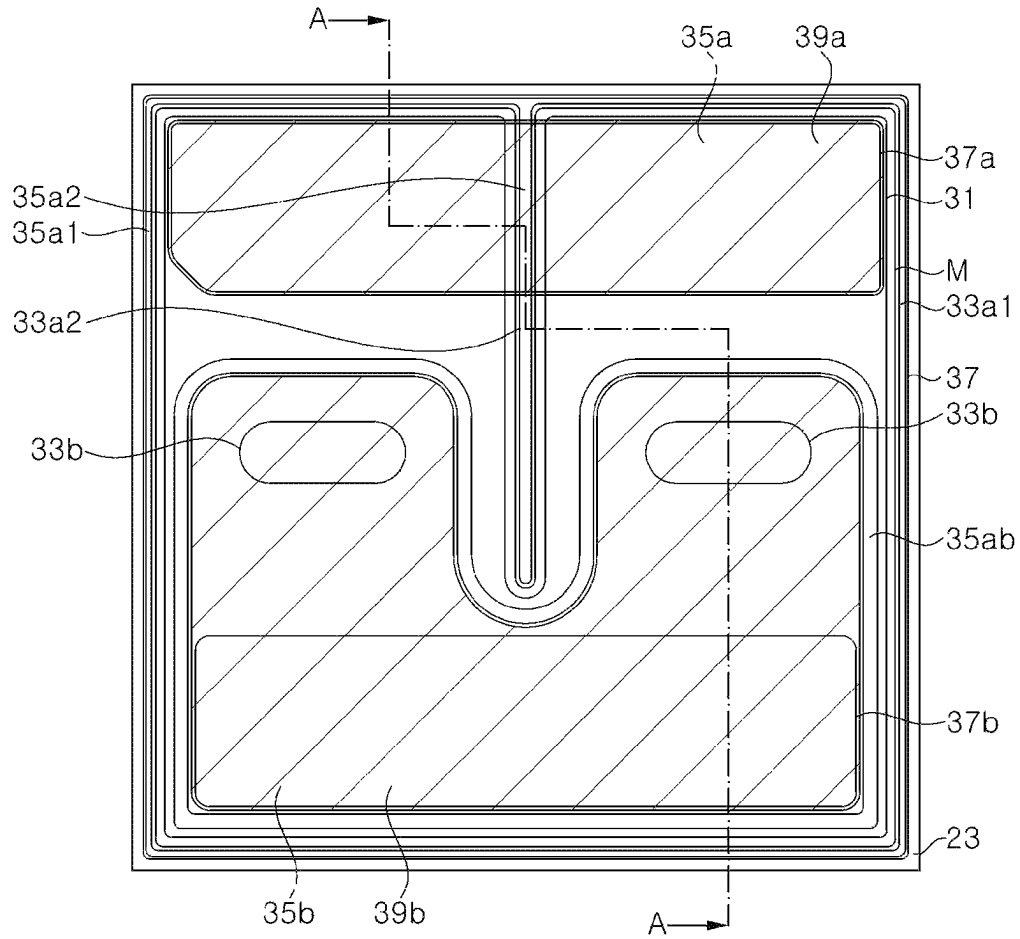


FIG. 4

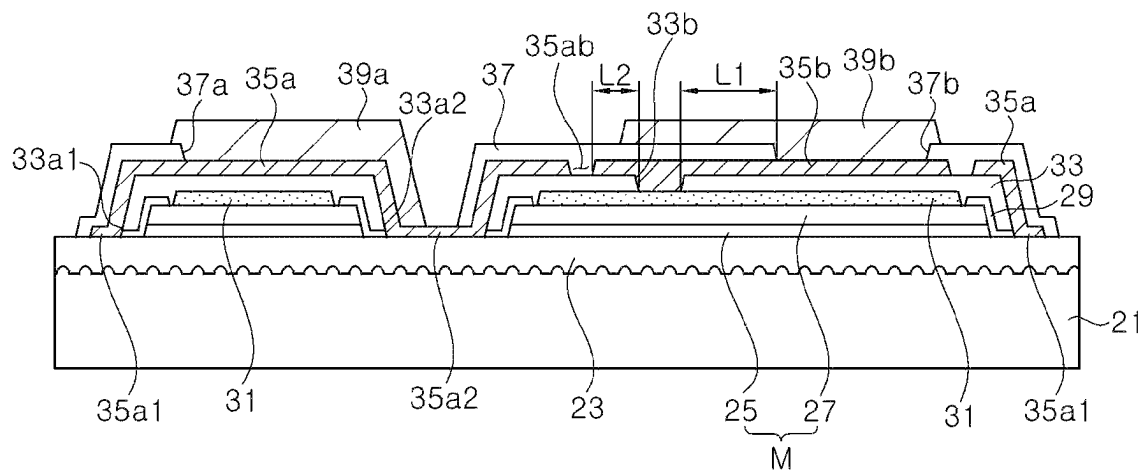


FIG. 5

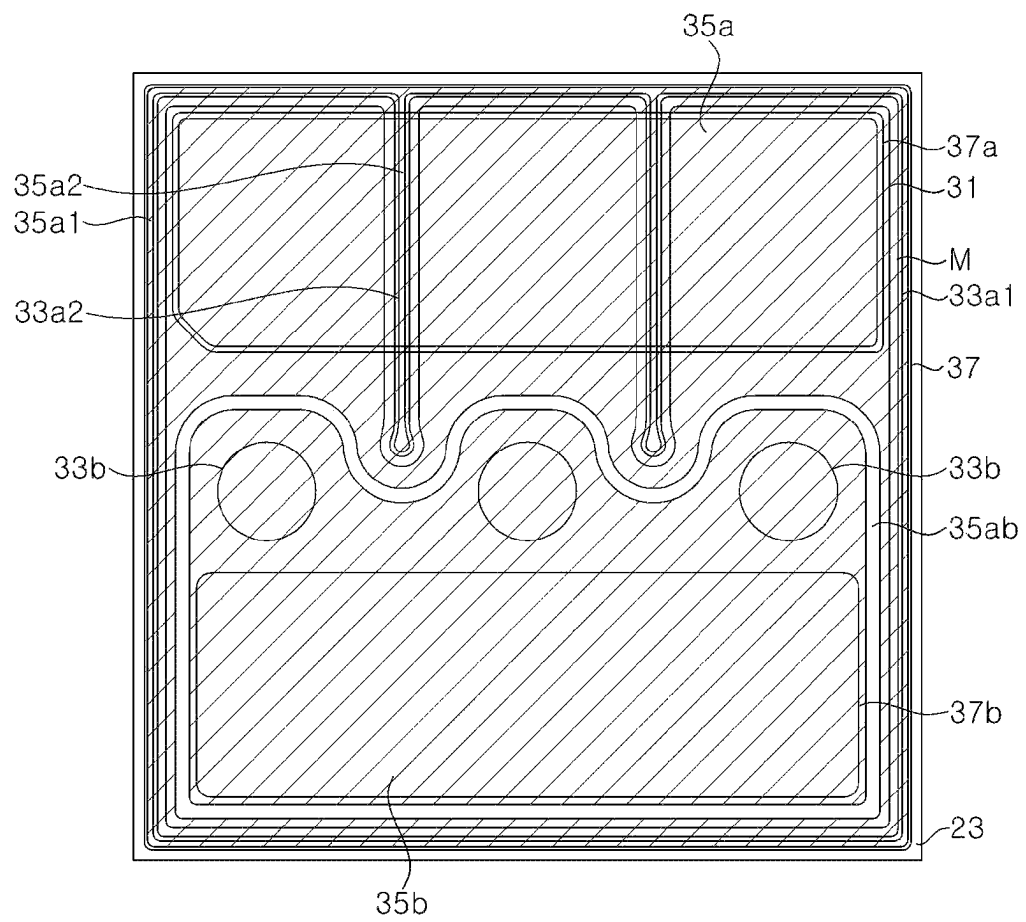


FIG. 6

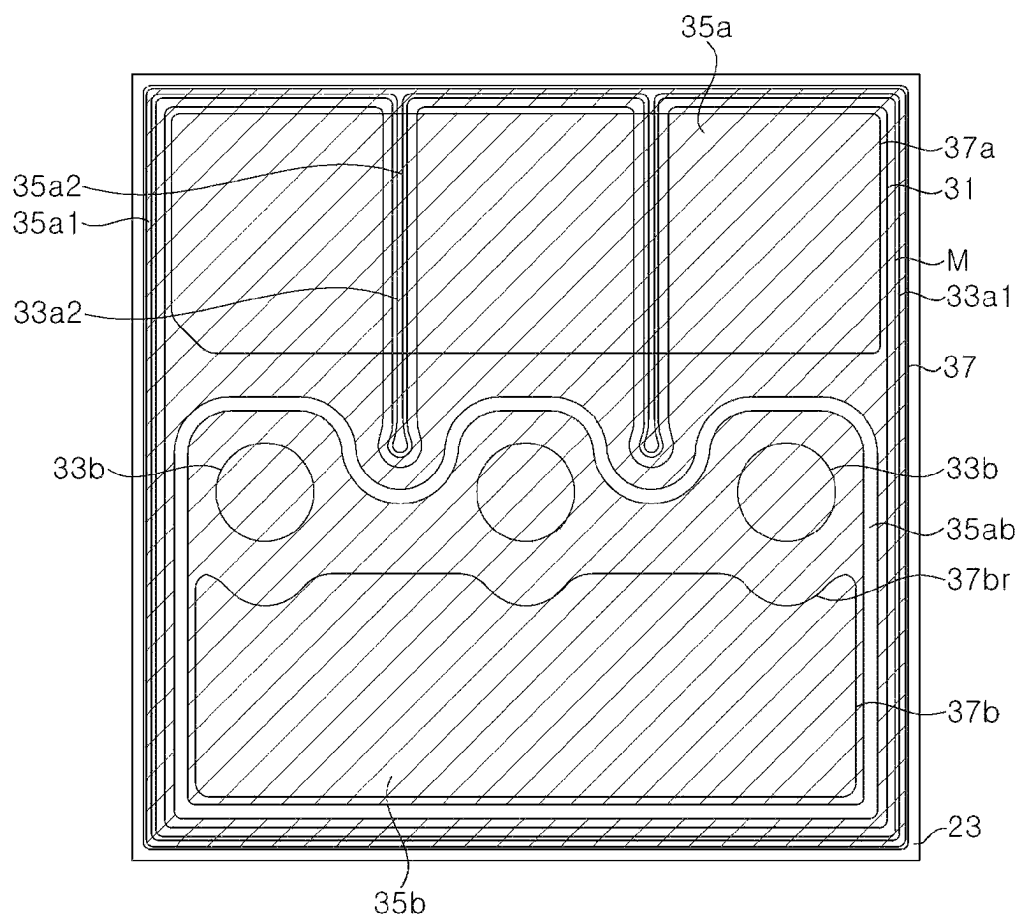


FIG. 7

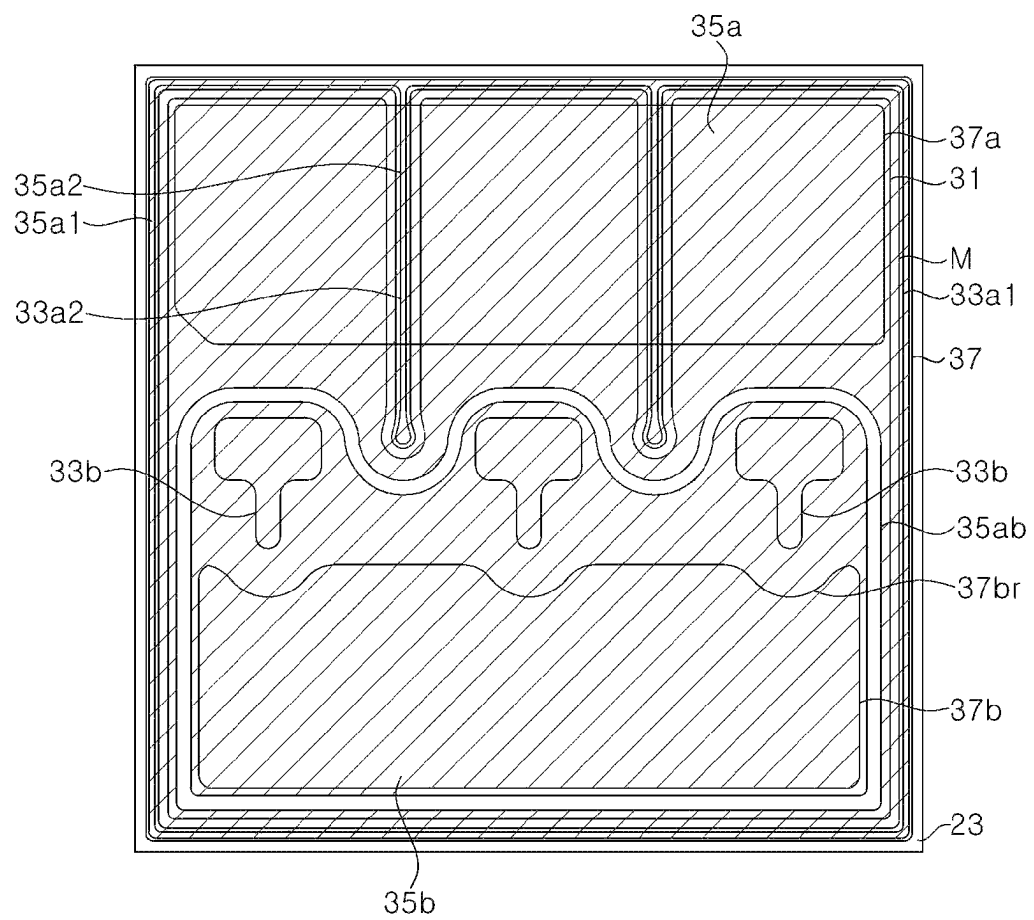


FIG. 8

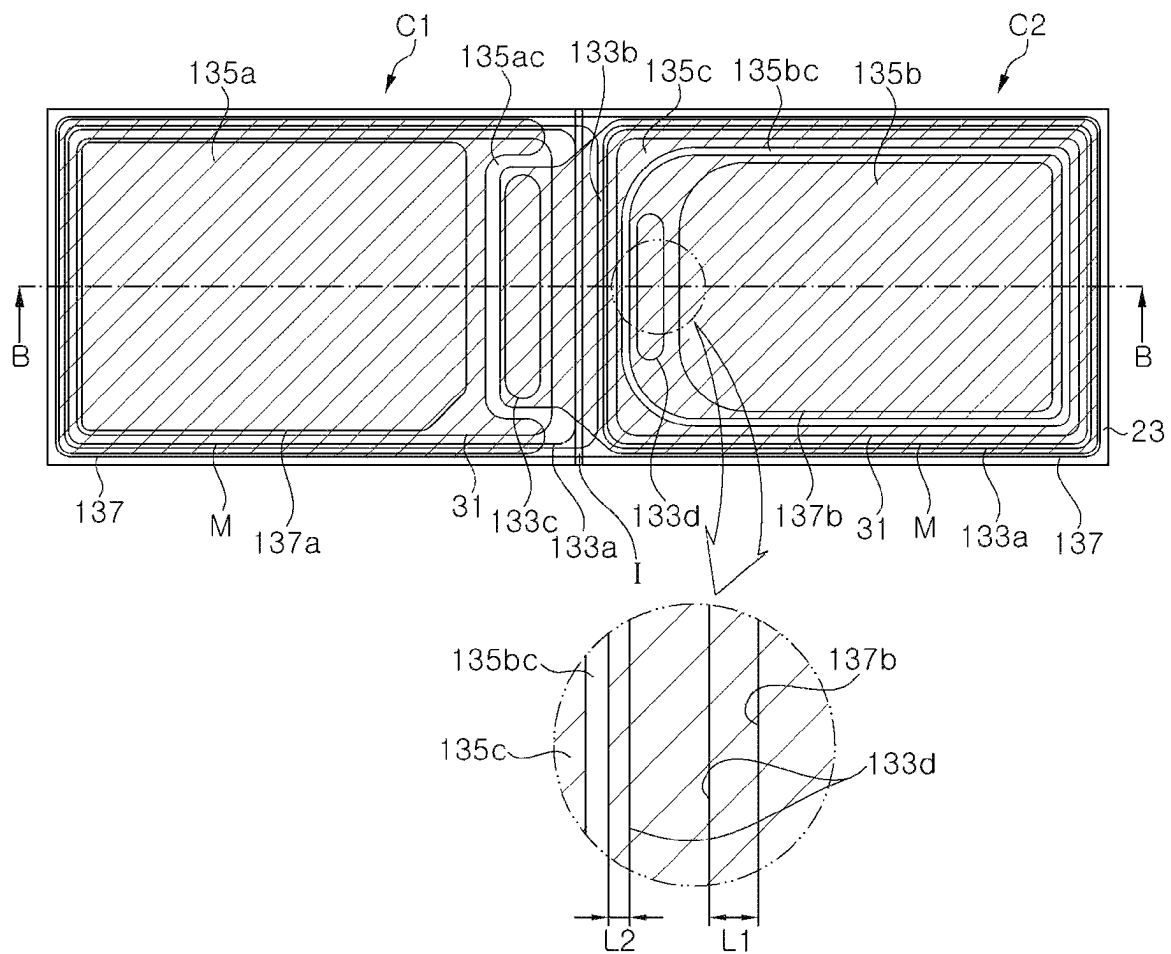




FIG. 9

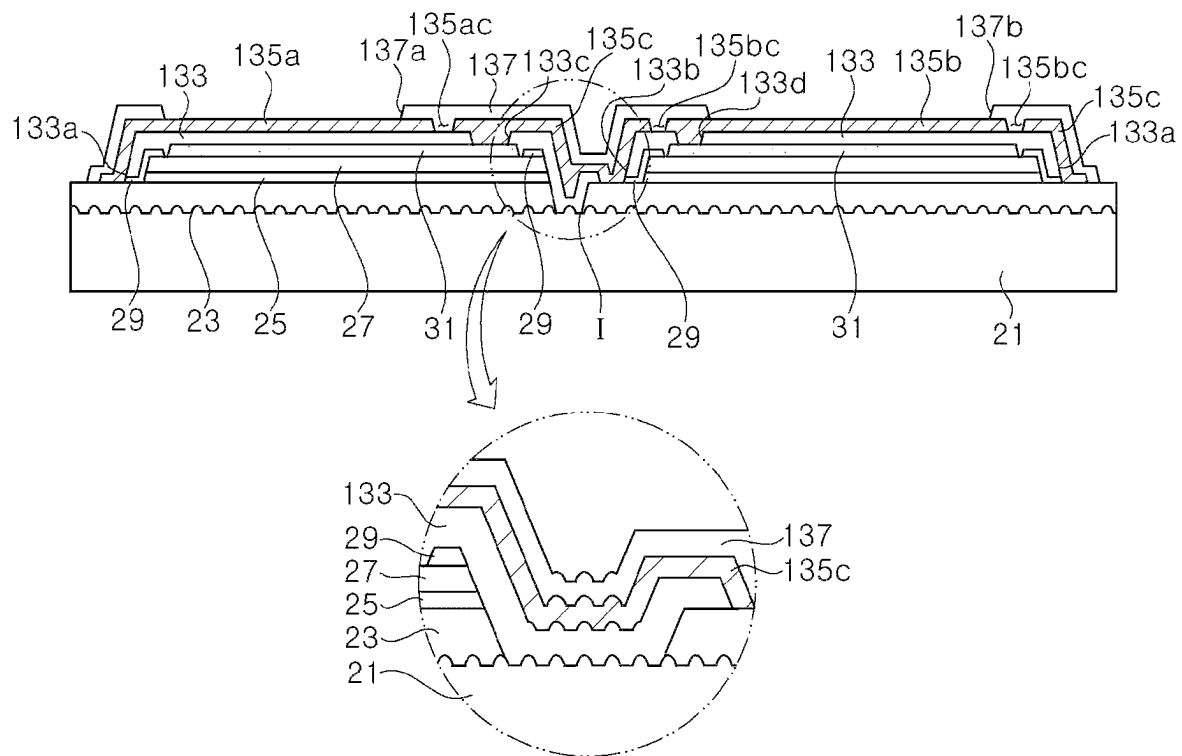


FIG. 10

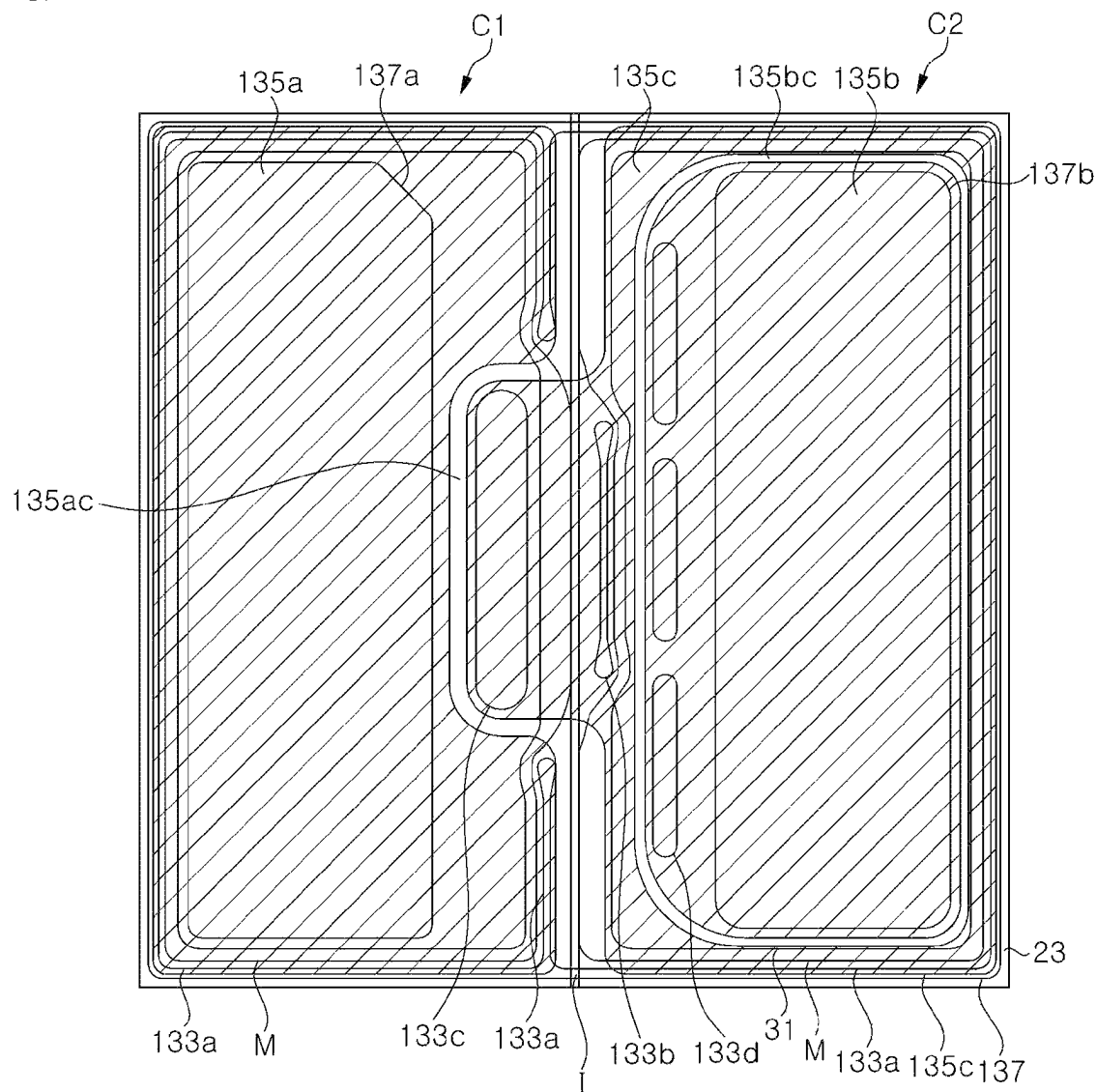


FIG. 11

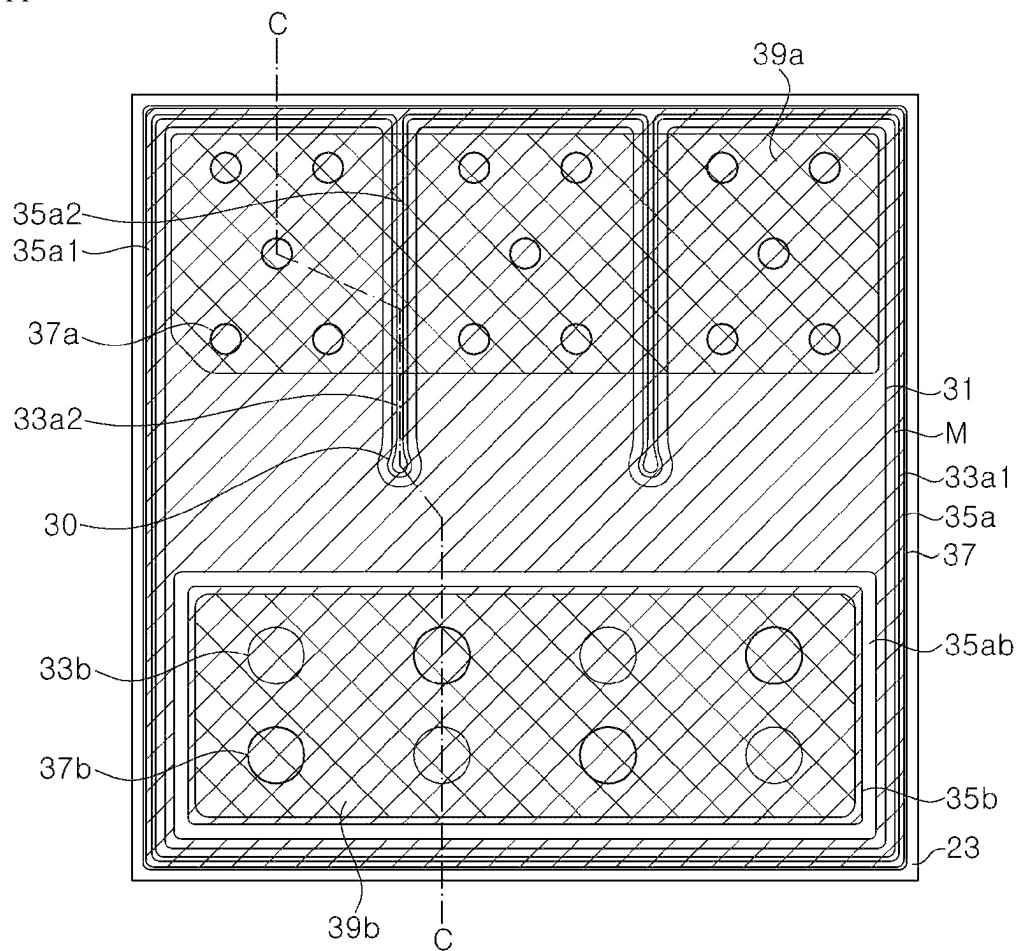


FIG. 12

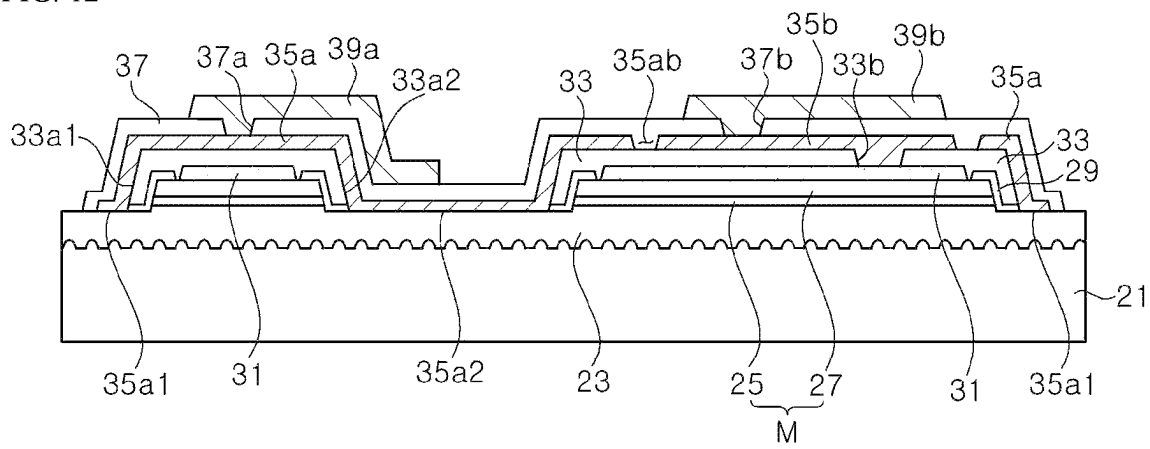


FIG. 13A

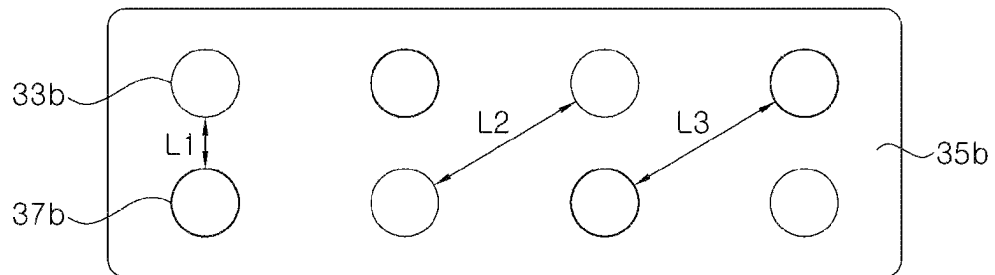


FIG. 13B

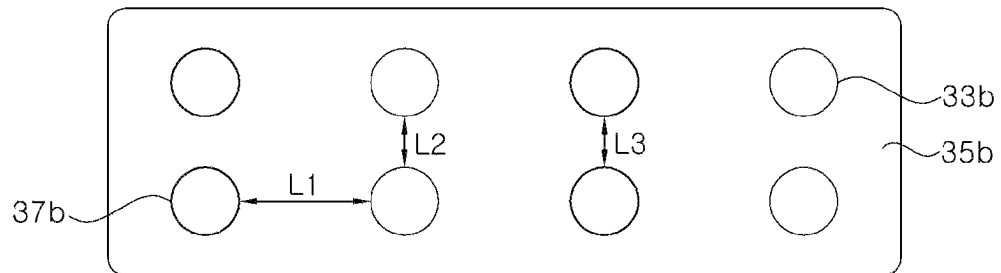


FIG. 14

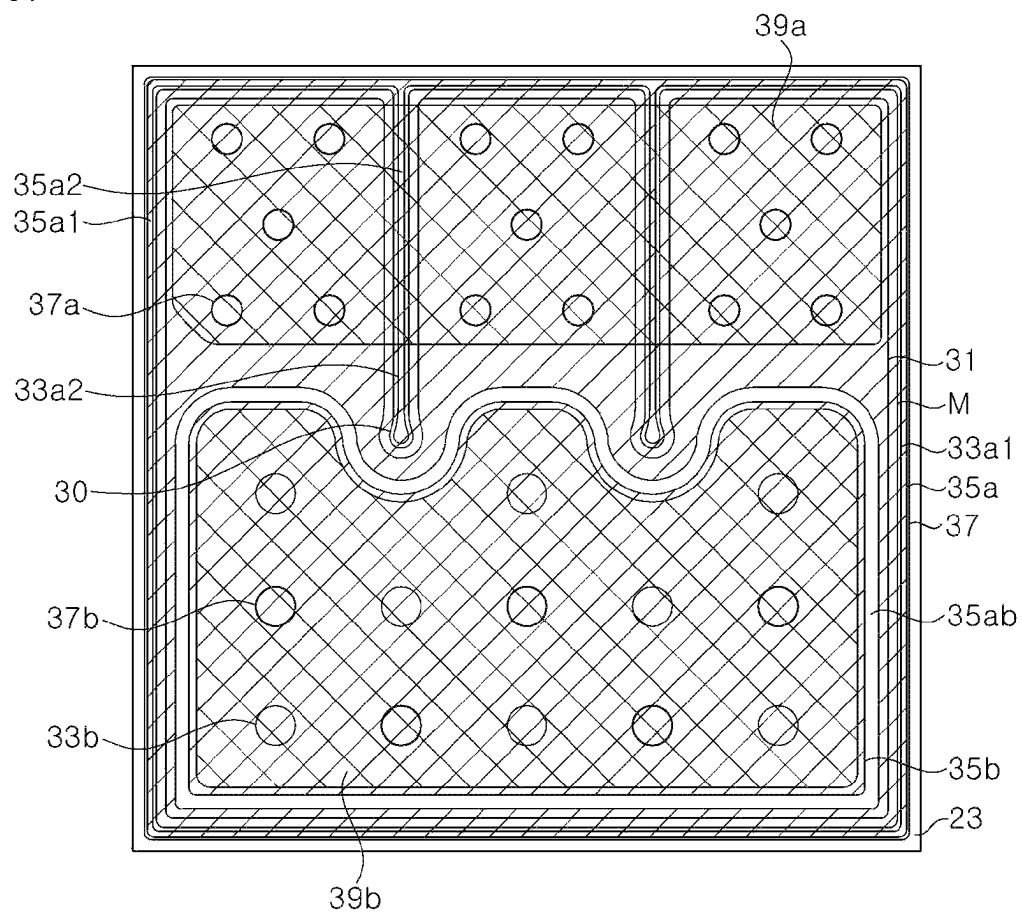


FIG. 15

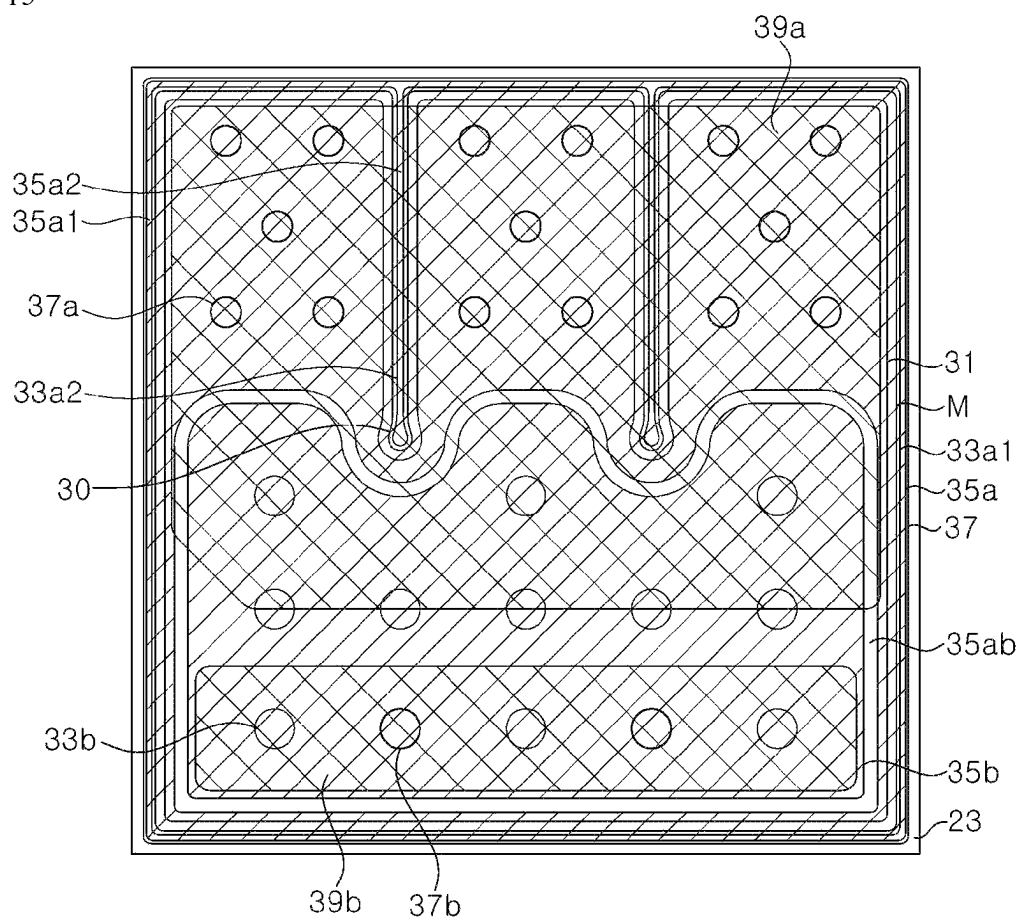




FIG. 18

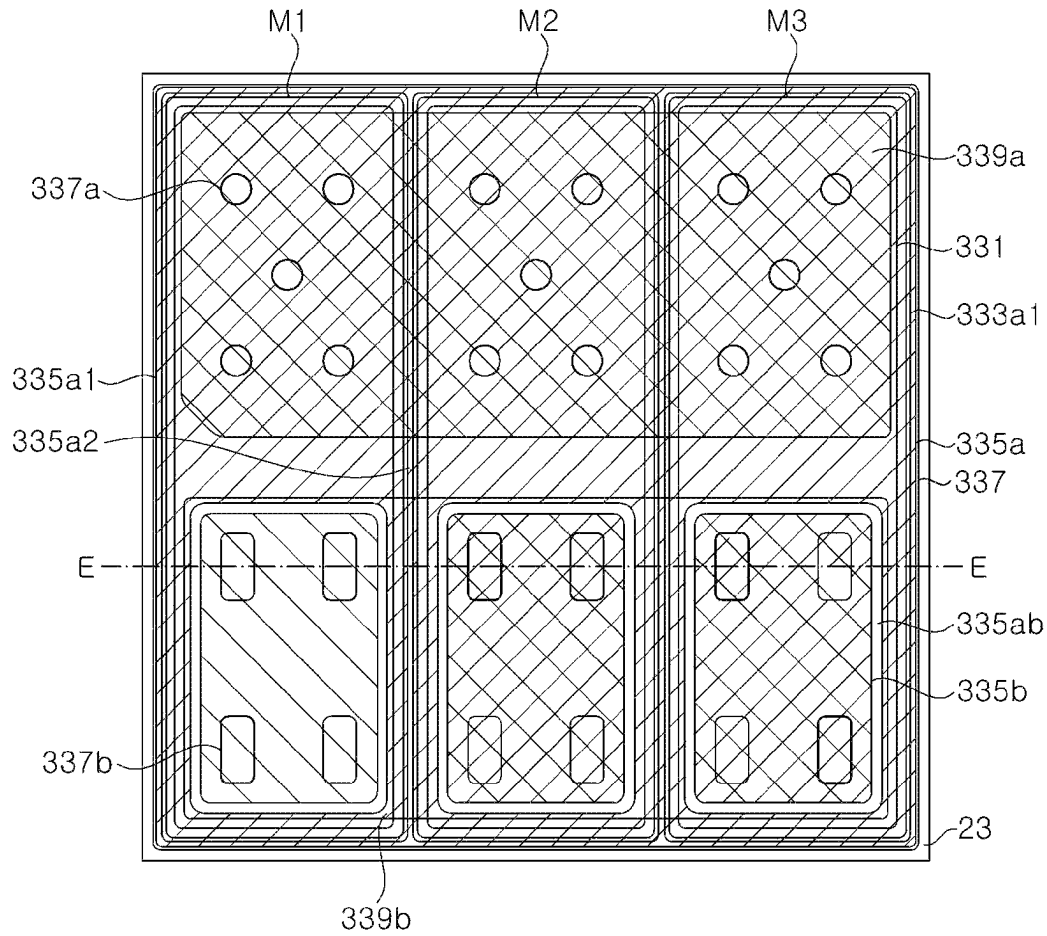


FIG. 19

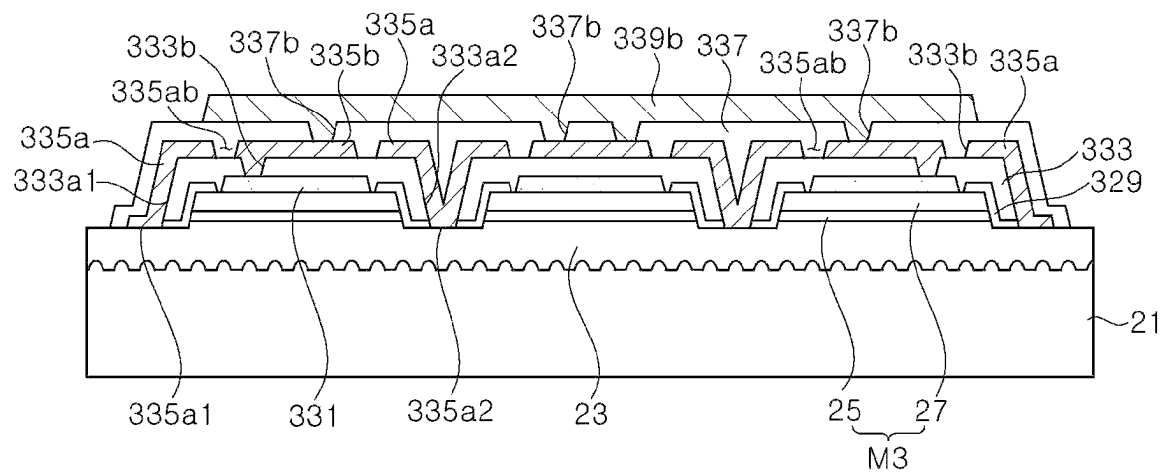




FIG. 20A

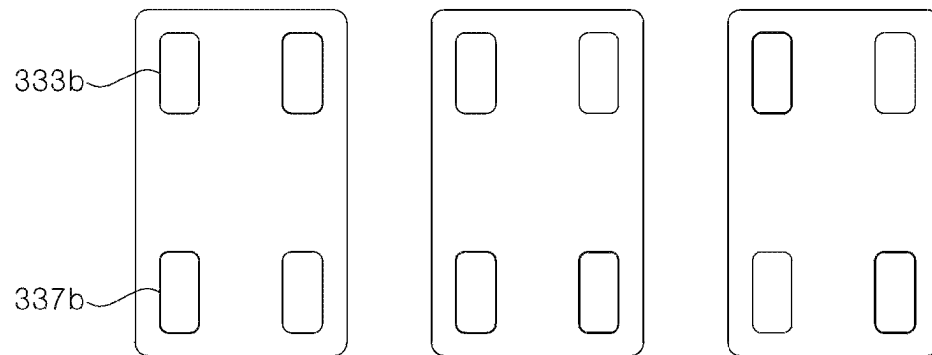


FIG. 20B

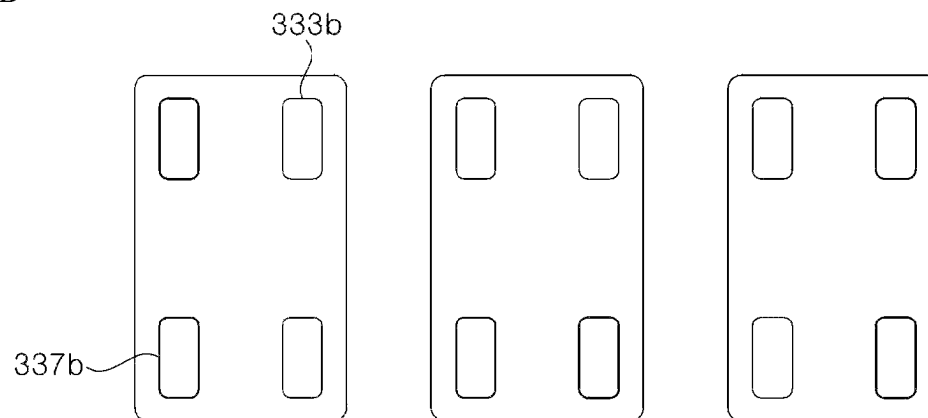


FIG. 21

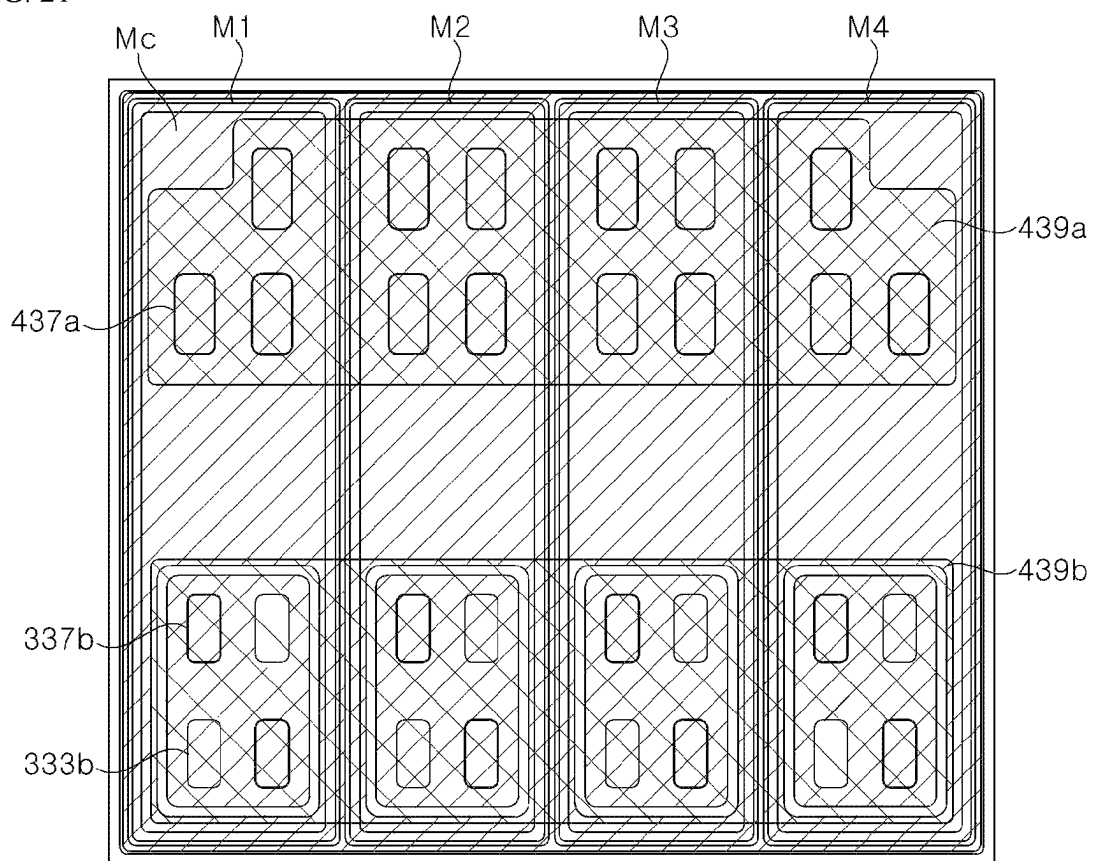


FIG. 22

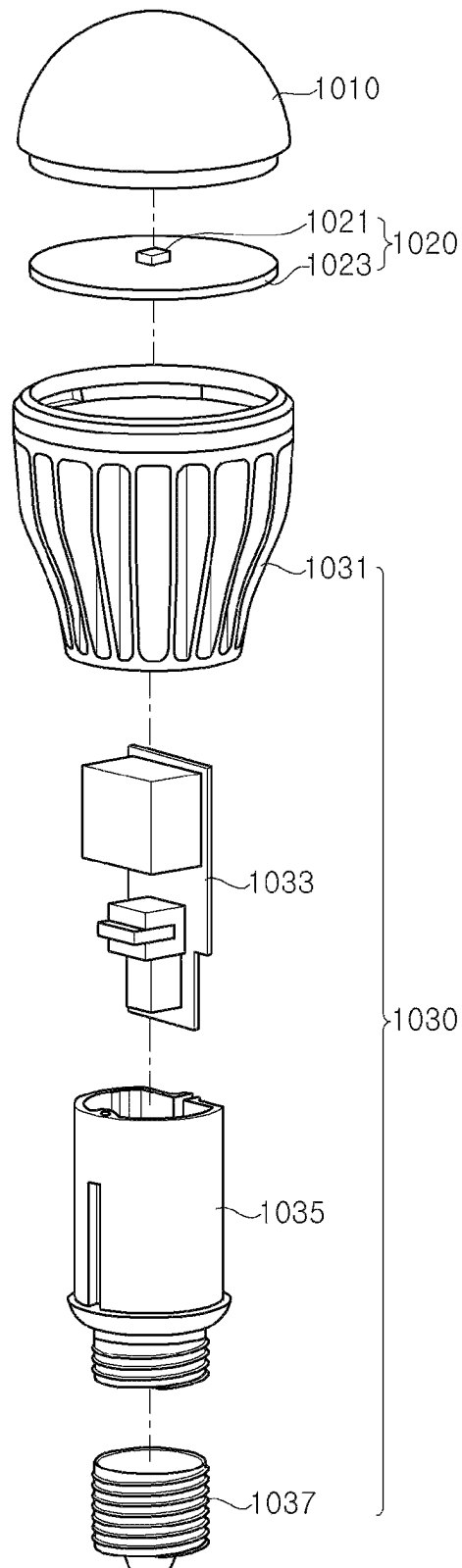


FIG. 23

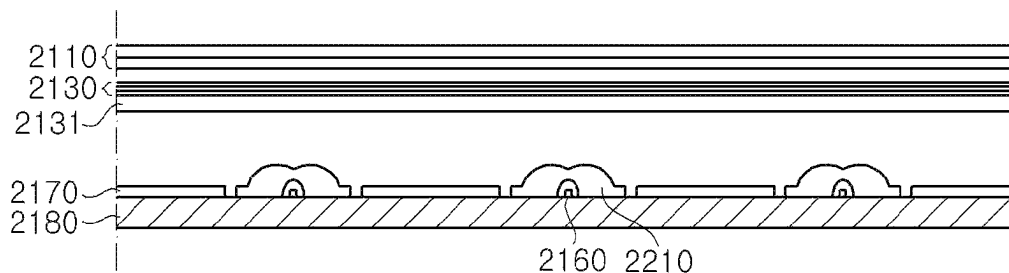


FIG. 24

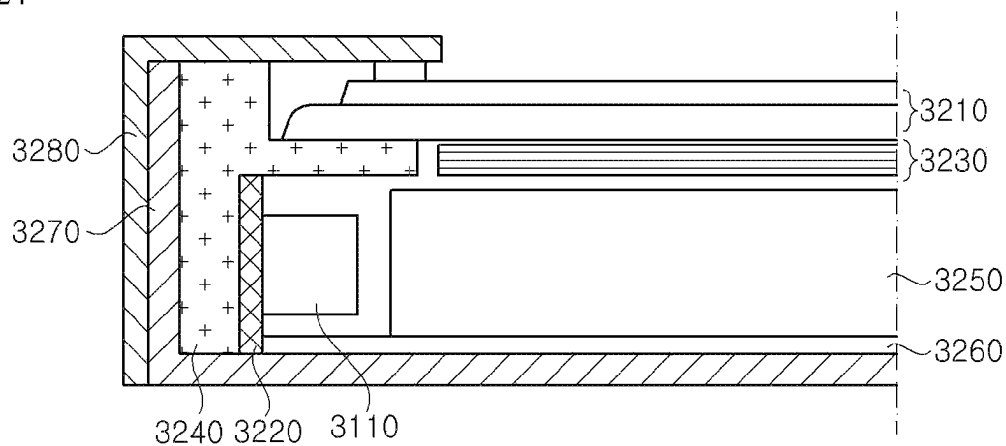
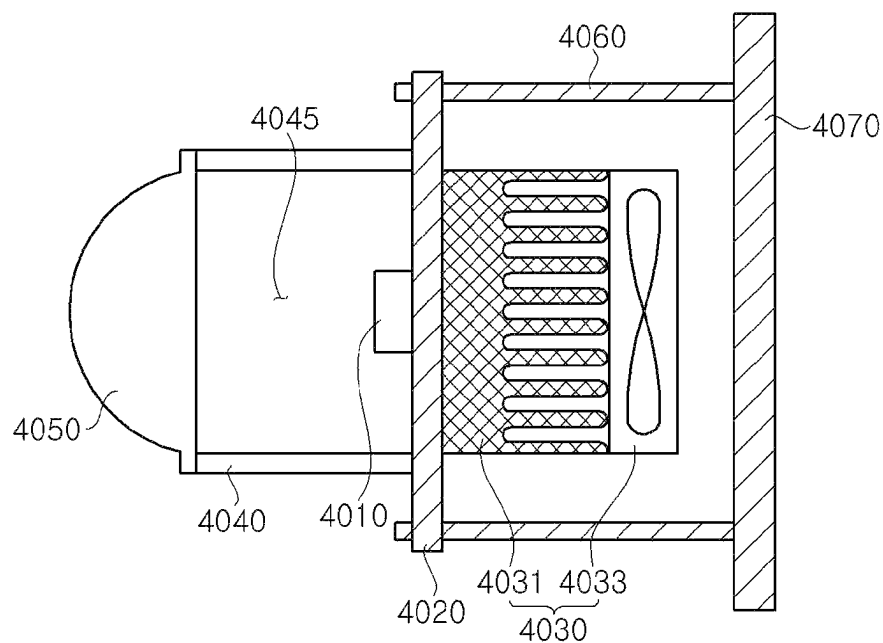


FIG. 25



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**CHIP-SCALE PACKAGE LIGHT EMITTING DIODE****CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a Continuation of U.S. patent application Ser. No. 17/163,629, filed on Feb. 1, 2021, which is a Divisional of U.S. patent application Ser. No. 16/284,468, filed on Feb. 25, 2019, now issued as U.S. Pat. No. 10,985,206, which is a Continuation of International Patent Application No. PCT/KR2017/009562, filed on Aug. 31, 2017, and claims priority to and benefit of Korean Patent Application No. 10-2016-0114057, filed on Sep. 5, 2016, and Korean Patent Application No. 10-2016-0147563, filed on Nov. 7, 2016, each of which is hereby incorporated by reference for all purposes as if fully set forth herein.

**BACKGROUND****Field**

Exemplary embodiments of the inventive concepts relate to a light emitting diode, more specifically to a chip-scale package type light emitting diode.

**Discussion of the Background**

In general, with good thermal stability and a direct transition type energy band structure, Group III-based nitrides, such as gallium nitride (GaN), aluminum nitride (AlN), and the like, have been identified as materials for light sources in the visible range and the ultraviolet range. In particular, blue and green light emitting diodes using indium gallium nitride are used in various fields including large full color flat panel displays, signal lamps, interior lighting, high density light sources, high resolution output systems, optical communication, and the like.

In recent years, research has been conducted on a chip-scale package type light emitting diode in which a packaging process is performed at a chip level. Since these light emitting diodes are smaller in size than standard packages and do not require a separate packaging process, manufacturing processes can further be simplified, and time and cost can be saved.

The chip-scale package type light emitting diode has a flip-chip shape electrode structure in general, and thus has excellent heat dissipation characteristics. However, since the chip-scale package type light emitting diode is generally manufactured to have the flip-chip shape electrode structure, there is a problem that a structure of the light emitting diode is considerably complicated in order to prevent diffusion of solder used in flip bonding. Nevertheless, solder, particularly Sn, may diffuse into the light emitting diode to contaminate an ohmic reflection layer and cause a failure of the light emitting diode.

Accordingly, it would be desirable to provide a reliable light emitting diode while simplifying the structure of the light emitting diode.

The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

**SUMMARY**

Exemplary embodiments of the inventive concepts provide a light emitting diode capable of effectively preventing

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diffusion of a bonding material such as solder without complicating a structure of the light emitting diode, thereby improving reliability.

Exemplary embodiments of the inventive concepts provide a light emitting diode capable of effectively preventing diffusion of the solder within a limited design range.

Exemplary embodiments of the inventive concepts provide a light emitting diode capable of preventing diffusion of the solder in the light emitting diode including a plurality of light emitting cells.

In accordance with one exemplary embodiment of the inventive concepts, a light emitting diode comprises: a substrate; a first conductivity type semiconductor layer disposed on the substrate; a mesa disposed on the first conductivity type semiconductor layer, and including an active layer and a second conductivity type semiconductor layer; an ohmic reflection layer disposed on the mesa and electrically connected to the second conductivity type semiconductor layer; a lower insulation layer covering the mesa and the ohmic reflection layer, and including a first opening exposing the first conductivity type semiconductor layer and a second opening exposing the ohmic reflection layer; a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer through the first opening; a second pad metal layer disposed on the lower insulation layer, and electrically connected to the ohmic reflection layer through the second opening; and an upper insulation layer covering the first pad metal layer and the second pad metal layer, and including a first opening exposing the first pad metal layer and a second opening exposing the second pad metal layer, wherein the second opening of the upper insulation layer is separated from the second opening of the lower insulation layer.

Since the second opening of the upper insulation layer is separated from the second opening of the lower insulation layer, it is possible to prevent the solder from diffusing into the ohmic reflection layer.

Furthermore, a shortest distance from the second opening of the lower insulation layer to the second opening of the upper insulation layer may be greater than a shortest distance from the second opening of the lower insulation layer to an edge of the second pad metal layer. The upper insulation layer and the lower insulation layer prevent diffusion of the solder, but the solder may reach the second opening of the lower insulation layer along an interface of the lower insulation layer and the second pad metal layer. Therefore, by spacing the second opening of the lower insulation layer away from the second opening of the upper insulation layer within the limited design range, a diffusion path of the solder may be increased, thereby preventing defects due to diffusion of the solder.

Meanwhile, the first opening of the lower insulation layer may expose the first conductivity type semiconductor layer along a periphery of the mesa, and the first pad metal layer may have an outer contact portion contacting the first conductivity type semiconductor layer along the periphery of the mesa. The first pad metal layer contacts the first conductivity type semiconductor layer along the periphery of the mesa, so that current spreading capability of the light emitting diode may be improved.

In addition, the mesa may comprise an indent portion to expose the first conductivity type semiconductor layer, and the first opening of the lower insulation layer may further expose the first conductivity type semiconductor layer in the indent portion. Furthermore, the first pad metal layer may further comprise an inner contact portion contacting the first

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conductivity type semiconductor layer in the indent portion. Since the first pad metal layer contacts the first conductivity type semiconductor layer at the periphery of the mesa and inside the mesa, current spreading capability of the light emitting diode is further enhanced.

Furthermore, the inner contact portion may be connected to the outer contact portion, but the present disclosure is not limited thereto, the inner contact portion and the outer contact portion may be separated from each other.

Meanwhile, the second opening of the lower insulation layer may have a convex shape on a side facing the second opening of the upper insulation layer, and the second opening of the upper insulation layer may have a concave shape corresponding to the convex shape of the second opening of the lower insulation. The second opening of the lower insulation layer may have a smaller size than the second opening of the upper insulation layer. Therefore, it is advantageous for patterning that the lower insulation layer has the convex shape. In addition, the second opening of the upper insulation layer has the concave portion so that a distance from the second opening of the lower insulation layer to the second opening of the upper insulation layer may be increased.

In some embodiments, the first and second pad metal layers exposed through the first and second openings of the upper insulation layer may be bonding pads to which the solder is directly bonded. Therefore, the solder contacts upper surfaces of first and second pad electrode layers exposed through the openings of the upper insulation layer.

In other embodiments, the light emitting diode further comprises first and second bump pads that cover the first and second pad metal layers exposed through the first and second openings of the upper insulation layer, respectively. By adopting the first and second bump pads, the diffusion path of the solder may be made long.

Furthermore, the first and second bump pads may cover and seal the first and second openings of the upper insulation layer, respectively. Accordingly, it is possible to prevent the solder from directly contacting the first and second pad metal layers.

Meanwhile, the second bump pad may also cover an upper insulation layer on an upper portion of the second opening of the lower insulation layer.

A light emitting diode in accordance with another exemplary embodiment comprises: a substrate; a first light emitting cell and a second light emitting cell disposed adjacent to each other on the substrate, and each comprising a first conductivity type semiconductor layer, a second conductivity type semiconductor layer, and an active layer disposed between the first conductivity type semiconductor layer and the second conductivity type semiconductor layer; ohmic reflection layers disposed on the second conductivity type semiconductor layers of the first light emitting cell and the second light emitting cell, respectively; a lower insulation layer covering the first light emitting cell, the second light emitting cell and ohmic reflection layers, and including first openings for exposing the first conductivity type semiconductor layers of the first and second light emitting cells and second openings for exposing the ohmic reflection layers; a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer of the first light emitting cell through the first opening on the first light emitting cell; a second pad metal layer disposed on the lower insulation layer and electrically connected to the ohmic reflection layer on the second light emitting cell through the second opening on the second light emitting cell; and an upper insulation layer

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having a first opening exposing the first pad metal layer and a second opening exposing the second pad metal layer, wherein the second opening of the upper insulation layer is separated from the second openings of the lower insulation layer.

Accordingly, it is possible to provide the light emitting diode capable of preventing diffusion of solder in the light emitting diode including a plurality of light emitting cells.

Furthermore, a shortest distance from the second opening of the lower insulation layer on the second light emitting cell to the second opening of the upper insulation layer on the second light emitting cell may be greater than a shortest distance from the second opening of the lower insulation layer on the second light emitting cell to an edge of the second pad metal layer. Therefore, diffusion of the solder may be effectively prevented within a limited design range.

Meanwhile, the light emitting diode may further comprise a connection metal layer disposed on the lower insulation layer, and electrically connected to the ohmic reflection layer on the first light emitting cell through the second opening of the first light emitting cell, and electrically connected to the first conductivity type semiconductor layer of the second light emitting cell through the first opening of the second light emitting cell. Accordingly, the light emitting cells may be connected to each other in series.

Furthermore, the first openings of the lower insulation layer may expose the first conductivity type semiconductor layer on at least one side of each light emitting cell along an edge of each of the first and second light emitting cells, the first pad metal layer may contact the first conductivity type semiconductor layer along the edge of the first light emitting cell, and the connection metal layer may contact the first conductivity type semiconductor layer along the edge of the second light emitting cell. A region where the first pad metal layer and the connection metal layer contact the first conductivity type semiconductor layer may be continuous or intermittent.

Moreover, the connection metal layer may contact the first conductivity type semiconductor layer on at least one side along the edge of the second light emitting cell. In particular, the connection metal layer may contact the first conductivity type semiconductor layer on all four sides along the edge of the second light emitting cell. Accordingly, current spreading capability is improved in the second light emitting cell.

The second opening of the lower insulation layer on the second light emitting cell may be one, but the present disclosure is not limited thereto, the second light emitting cell may be plural.

In some embodiments, the first and second pad metal layers exposed through the first and second openings of the upper insulation layer may be bonding pads to which the solder is directly bonded. Therefore, the solder contacts upper surfaces of first and second pad electrode layers exposed through the openings of the upper insulation layer.

In other embodiments, the light emitting diode further comprises first and second bump pads that cover the first and second pad metal layers exposed through the first and second openings of the upper insulation layer, respectively.

The first and second bump pads may cover and seal the first and second openings of the upper insulation layer, respectively.

Furthermore, the second bump pad may also cover an upper insulation layer on an upper portion of the second opening of the lower insulation layer.

In accordance with another exemplary embodiment of the inventive concepts, a light emitting diode comprises: a first conductivity type semiconductor layer; a mesa disposed on

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the first conductivity type semiconductor layer, and including an active layer and a second conductivity type semiconductor layer; an ohmic reflection layer disposed on the mesa and electrically connected to the second conductivity type semiconductor layer; a lower insulation layer covering the mesa and the ohmic reflection layer, and including a first opening exposing the first conductivity type semiconductor layer and a second opening exposing the ohmic reflection layer; a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer through the first opening; a second pad metal layer disposed on the lower insulation layer, and electrically connected to the ohmic reflection layer through the second opening; and an upper insulation layer covering the first pad metal layer and the second pad metal layer, and including a first opening exposing the first pad metal layer and a plurality of second openings exposing the second pad metal layer, wherein the second openings of the upper insulation layer are separated from the second opening of the lower insulation layer not to overlap each other.

By forming the plurality of second openings in the upper insulation layer, a diffusion path of the solder may be reduced, and furthermore, the plurality of second openings in the upper insulation layer is separated from the second opening in the lower insulation layer so that it is possible to prevent solder from diffusing to the reflection layer.

Meanwhile, the light emitting diode may further include: a first bump pad connected to the first pad metal layer through the first opening of the upper insulation layer; and a second bump pad connected to the second pad metal layer through the plurality of second openings of the upper insulation layer. In addition, the first conductivity type semiconductor layer may be disposed on a substrate.

In some embodiments, a shortest distance from the second opening of the lower insulation layer to the second opening of the upper insulation layer may be greater than a shortest distance between the second openings of the upper insulation layer.

In some embodiments, the lower insulation layer may comprise a plurality of second openings, and a shortest distance from the second opening of the lower insulation layer to the second opening of the upper insulation layer may be greater than a shortest distance between the second openings of the lower insulation layer.

The upper insulation layer and the lower insulation layer prevent the solder from diffusing, but the solder may reach the second opening of the lower insulation layer along an interface between the lower insulation layer and the second pad metal layer. Therefore, by spacing the second opening of the lower insulation layer away from the second opening of the upper insulation layer within a limited design range, the diffusion path of the solder may be increased, thereby preventing defects due to diffusion of the solder.

The first opening of the lower insulation layer may expose the first conductivity type semiconductor layer along a periphery of the mesa, and the first pad metal layer may have an outer contact portion contacting the first conductivity type semiconductor layer along the periphery of the mesa. The first pad metal layer contacts the first conductivity type semiconductor layer along the periphery of the mesa, so that current spreading capability of the light emitting diode may be improved.

In addition, the mesa may comprise an indent portion to expose the first conductivity type semiconductor layer, and the first opening of the lower insulation layer may further expose the first conductivity type semiconductor layer in the indent portion. Furthermore, the first pad metal layer may

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further comprise an inner contact portion contacting the first conductivity type semiconductor layer in the indent portion. Since the first pad metal layer contacts the first conductivity type semiconductor layer at the periphery of the mesa and inside the mesa, current spreading capability of the light emitting diode is further enhanced.

Furthermore, the inner contact portion may be connected to the outer contact portion, but the present disclosure is not limited thereto, the inner contact portion and the outer contact portion may be separated from each other.

In some embodiments, the mesa may have a via hole exposing the first conductivity type semiconductor layer through the second conductivity type semiconductor layer and the active layer, wherein the first opening of the lower insulation layer may expose the first conductivity type semiconductor layer exposed in the via hole, and the first pad metal layer may have an inner contact portion contacting the first conductivity type semiconductor layer exposed in the via hole.

Moreover, the first pad metal layer may comprise outer contact portions contacting the first conductivity type semiconductor layer outside the mesa, wherein the outer contact portions may be spaced apart from one another.

Meanwhile, the lower insulation layer may comprise the plurality of second openings, and the second bump pad may cover an upper portion of at least one second opening of the lower insulation layer. Furthermore, the second bump pad may cover entire upper portions of the second openings of the lower insulation layer.

Furthermore, the first and second bump pads may cover and seal the first and second openings of the upper insulation layer, respectively. The first and second bump pads prevent the first and second pad metal layers from being exposed to the solder. In addition, by forming the plurality of the first and second openings of the upper insulation layer, it is possible to reduce the diffusion path of the solder diffused into the first and second pad metal layers through the first and second bump pads, thereby delaying the diffusion of the solder.

Meanwhile, the first bump pad may cover the upper portion of at least one second opening of the lower insulation layer. Any location and shape of the first bump pad capable of being insulated from the second pad metal layer may be variously changed, and any location and shape of the second bump pad capable of being insulated from the first pad metal layer may also be variously changed. For example, the second bump pad may comprise a protrusion between the first bump pad and the second bump pad. Further, at least one of the second openings of the lower insulation layer may be disposed under the protrusion.

Meanwhile, the second pad metal layer may be surrounded by the first pad metal layer. Accordingly, a boundary region in which the lower insulation layer is exposed may be formed between the first pad metal layer and the second pad metal layer. This boundary region may be covered by the upper insulation layer.

In some embodiments, the lower insulation layer may comprise a plurality of second openings, and at least one of the second openings of the upper insulation layer may be disposed between the two second openings of the lower insulation layer.

Meanwhile, a plurality of mesas may be disposed on the first conductivity type semiconductor layer, the second opening of the lower insulation layer and second openings of the upper insulation layer may be disposed on each mesa, and each of the bump pad and the second bump pad may be disposed over the plurality of mesas. Further, the first pad

metal layer may cover the mesas, and the second pad metal layer may be disposed on each mesa.

In some embodiments, the second bump pad may be disposed within an upper region of the second pad metal layer, but the inventive concepts are not limited thereto, the second bump pad may partially overlap with the first pad metal layer.

In accordance with embodiments of the inventive concepts, the opening of the upper insulation layer is spaced apart from the opening of the lower insulation layer to prevent the solder, particularly Sn, from diffusing into the ohmic reflection layer through the opening of the lower insulation layer, thereby improving a reliability of the light emitting diode. The diffusion of the solder may be effectively blocked within the limited design range by setting the shortest distance from the opening of the upper insulation layer to the opening of the lower insulation layer to be greater than the shortest distance from the opening of the lower insulation layer to the pad metal layer.

Other advantages and effects of the inventive concepts will become more apparent from the detailed description.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate exemplary embodiments of the invention, and together with the description serve to explain the inventive concepts.

FIG. 1 is a schematic plan view illustrating a light emitting diode according to an exemplary embodiment of the inventive concepts.

FIG. 2 is a cross-sectional view taken along the line A-A in FIG. 1.

FIG. 3 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 4 is a cross-sectional view taken along the line A-A of FIG. 3.

FIG. 5 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 6 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 7 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 8 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 9 is a cross-sectional view taken along the line B-B of FIG. 8.

FIG. 10 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 11 is a schematic plan view illustrating a light emitting diode according to an exemplary embodiment of the inventive concepts.

FIG. 12 is a cross-sectional view taken along the line C-C of FIG. 11.

FIGS. 13A and 13B are schematic plan views illustrating openings of a lower insulation layer and an upper insulation layer in relation to the embodiment of FIG. 11.

FIG. 14 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 15 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 16 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 17 is a cross-sectional view taken along the line D-D of FIG. 16.

FIG. 18 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 19 is a cross-sectional view taken along the line E-E of FIG. 19.

FIGS. 20A and 20B are schematic plan views illustrating openings of a lower insulation layer and an upper insulation layer in relation to the embodiment of FIG. 18.

FIG. 21 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

FIG. 22 is an exploded perspective view illustrating a lighting apparatus to which a light emitting diode according to an exemplary embodiment of the inventive concepts is applied.

FIG. 23 is a cross-sectional view illustrating a display apparatus to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

FIG. 24 is a cross-sectional view illustrating a display apparatus to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

FIG. 25 is a cross-sectional view illustrating a head light to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

#### DETAILED DESCRIPTION

In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various exemplary embodiments or implementations of the invention. As used herein “embodiments” and “implementations” are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various exemplary embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring various exemplary embodiments. Further, various exemplary embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an exemplary embodiment may be used or implemented in another exemplary embodiment without departing from the scope of the inventive concepts.

Unless otherwise specified, the illustrated exemplary embodiments are to be understood as providing exemplary features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as “elements”), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the scope of the inventive concepts.

The use of cross-hatching and/or shading in the accompanying drawings is generally provided to clarify boundaries



between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an exemplary embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals denote like elements.

When an element, such as a layer, is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z-axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, “at least one of X, Y, and Z” and “at least one selected from the group consisting of X, Y, and Z” may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

Although the terms “first,” “second,” etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

Spatially relative terms, such as “beneath,” “below,” “under,” “lower,” “above,” “upper,” “over,” “higher,” “side” (e.g., as in “sidewall”), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms

“comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

Various exemplary embodiments are described herein with reference to sectional and/or exploded illustrations that are schematic illustrations of idealized exemplary embodiments and/or intermediate structures. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, exemplary embodiments disclosed herein should not necessarily be construed as limited to the particular illustrated shapes of regions, but are to include deviations in shapes that result from, for instance, manufacturing. In this manner, regions illustrated in the drawings may be schematic in nature and the shapes of these regions may not reflect actual shapes of regions of a device and, as such, are not necessarily intended to be limiting.

As is customary in the field, some exemplary embodiments are described and illustrated in the accompanying drawings in terms of functional blocks, units, and/or modules, such as control boards and control units. Those skilled in the art will appreciate that these blocks, units, and/or modules are physically implemented by electronic (or optical) circuits, such as logic circuits, discrete components, microprocessors, hard-wired circuits, memory elements, wiring connections, and the like, which may be formed using semiconductor-based fabrication techniques or other manufacturing technologies. In the case of the blocks, units, and/or modules being implemented by microprocessors or other similar hardware, they may be programmed and controlled using software (e.g., microcode) to perform various functions discussed herein and may optionally be driven by firmware and/or software. It is also contemplated that each block, unit, and/or module may be implemented by dedicated hardware, or as a combination of dedicated hardware to perform some functions and a processor (e.g., one or more programmed microprocessors and associated circuitry) to perform other functions. Also, each block, unit, and/or module of some exemplary embodiments may be physically separated into two or more interacting and discrete blocks, units, and/or modules without departing from the scope of the inventive concepts. Further, the blocks, units, and/or modules of some exemplary embodiments may be physically combined into more complex blocks, units, and/or modules without departing from the scope of the inventive concepts.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

Hereinafter, exemplary embodiments of the inventive concepts will be described in detail with reference to the accompanying drawings. The following embodiments are

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provided by way of example so as to fully convey the spirit of the inventive concepts to those skilled in the art to which the present disclosure pertains. Accordingly, the present disclosure is not limited to the embodiments disclosed herein and can also be implemented in different forms. In the drawings, widths, lengths, thicknesses, and the like of elements can be exaggerated for clarity and descriptive purposes. When an element is referred to as being “disposed above” or “disposed on” another element, it can be directly “disposed above” or “disposed on” the other element, or intervening elements can be present. Throughout the specification, like reference numerals denote like elements having the same or similar functions.

FIG. 1 is a schematic plan view illustrating a light emitting diode according to an exemplary embodiment of the inventive concepts, and FIG. 2 is a cross-sectional view taken along the line A-A in FIG. 1.

Referring FIGS. 1 and 2, the light emitting diode comprises a substrate 21, a first conductivity type semiconductor layer 23, an active layer 25, a second conductivity type semiconductor layer 27, an ohmic reflection layer 31, a lower insulation layer 33, a first pad metal layer 35a, a second pad metal layer 35b, and an upper insulation layer 37. Further, the light emitting diode may further comprise a preliminary insulation layer 29.

The substrate 21 may be selected from any substrates suitable for growth of gallium nitride semiconductor layers thereon. The substrate 21 may include, for example, a sapphire substrate, a gallium nitride substrate, a SiC substrate, and the like, and may be a patterned sapphire substrate. The substrate 21 may have a rectangular or square shape, as shown in the plan view of FIG. 1, without being limited thereto. The size of the substrate 21 is not particularly limited and may be selected in various ways.

The first conductivity type semiconductor layer 23 is disposed on the substrate 21. The first conductivity type semiconductor layer 23 may be a gallium nitride semiconductor layer grown on the substrate 21 and doped with dopants, for example Si.

A mesa M is disposed on the first conductivity type semiconductor layer. The mesa M may be disposed within a region surrounded by the first conductivity type semiconductor layer 23 so that regions near edges of the first conductivity type semiconductor layer are not covered by the mesa M, but exposed to the outside.

The mesa M comprises the second conductivity type semiconductor layer 27 and the active layer 25. The active layer 25 is interposed between the first conductivity type semiconductor layer 23 and the second conductivity type semiconductor layer 27. The active layer 25 may have a single quantum well structure or a multiple quantum well structure. A composition and a thickness of the well layer in the active layer 25 determine a wavelength of generated light. In particular, by controlling the composition of the well layer, it is possible to provide an active layer generating ultraviolet light, blue light or green light.

Meanwhile, the second conductivity type semiconductor layer 27 may be a gallium nitride semiconductor layer doped with p-type dopants, for example, Mg. Each of the first conductivity type semiconductor layer 23 and the second conductivity type semiconductor layer 27 may be composed of a single layer or multiple layers and may include super-lattice layers. The first conductivity type semiconductor layer 23, the active layer 25 and the second conductivity type semiconductor layer 27 may be grown on the substrate in a chamber by a well-known method, such as metal organic chemical vapor deposition (MOCVD).

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As shown in FIG. 1, the mesa M may have an indent portion intruding therein, and an upper surface of the first conductivity type semiconductor layer 23 may be exposed by the indent portion. The indent portion may be formed long from one side edge of the mesa M toward the other side opposite to the mesa M, and may pass through a center of the mesa M. In the present embodiment, it is described that the mesa M has the indent portion, but the mesa M may have a rectangular shape similar to the substrate 21, and may be formed to have a through hole(s) exposing the first conductivity type semiconductor layer 23, for example.

Meanwhile, the ohmic reflection layer 31 is disposed over the mesa M and is electrically connected to the second conductivity type semiconductor layer 27. The ohmic reflection layer 31 may be disposed over substantially the entire region of the second conductivity type semiconductor layer 27 in an upper region of the mesa M. For example, the ohmic reflection layer 31 may cover 80% or more, further, 90% or more of the upper region of the mesa M.

The ohmic reflection layer 31 may include a reflective metallic layer and thus can reflect light generated from the active layer 25 and travelling to the ohmic reflection layer 31, toward the substrate 21. For example, the ohmic reflection layer 31 may be composed of a single reflective metallic layer, but is not limited thereto. Alternatively, the ohmic reflection layer 31 may include an ohmic layer and a reflection layer. The ohmic layer may be a metal layer such as a Ni layer or a transparent oxide layer such as an ITO layer, and the reflection layer may be a metal layer having high reflectance such as an Ag or Al layer.

Meanwhile, a preliminary insulation layer 29 may cover the mesa M in a periphery of the ohmic reflection layer 31. The preliminary insulation layer 29 may be formed of SiO<sub>2</sub>, and may cover a side of the mesa M and may further cover a partial region of the first conductivity type semiconductor layer 23. In another embodiment, the preliminary insulation layer 29 may be disposed only on the periphery of the ohmic reflection layer 31 in the upper region of the mesa M.

The lower insulation layer 33 covers the mesa M and the ohmic reflection layer 31. The lower insulation layer 33 may also cover the first conductivity type semiconductor layer 23 along a periphery of the mesa M, and may cover the first conductivity type semiconductor layer 23 in the indent portion inside the mesa M. The lower insulation layer 33 covers side surfaces of the mesa M in particular.

Meanwhile, the lower insulation layer 33 has first and second openings 33a1 and 33a2 exposing the first conductivity type semiconductor layer 23 and a second opening 33b exposing the ohmic reflection layer 31. The first opening 33a1 exposes the first conductivity type semiconductor layer 23 along the periphery of the mesa M, and the first opening 33a2 exposes the first conductivity type semiconductor layer 23 in the indent portion. As shown in FIG. 1, the first opening 33a1 and the first opening 33a2 may be connected to each other. However, the inventive concepts are not limited thereto, but the first opening 33a1 and the first opening 33a2 may be separated from each other.

The second opening 33b exposes the ohmic reflection layer 31. A plurality of second openings 33b may be formed, and these second openings 33b may be disposed on both sides of the indent portion. Locations of the second openings 33b will be described later.

Meanwhile, the lower insulation layer 33 covers the preliminary insulation layer 29 and is integrated with the preliminary insulation layer 29. Unless specifically stated, it should be understood that the preliminary insulation layer 29 is included in the lower insulation layer 33. The lower

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insulation layer 33 may be formed of a single layer of SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, without being limited thereto. For example, the lower insulation layer 33 may have a multilayer structure which includes a silicon oxide layer and a silicon nitride layer, and may include a distributed Bragg reflector in which silicon oxide layers and titanium oxide layers are alternately stacked one above another.

The first pad metal layer 35a is disposed on the lower insulation layer 33 and is insulated from the mesa M and the ohmic reflection layer 31 by the lower insulation layer 33. The first pad metal layer 35a contacts the first conductivity type semiconductor layer 23 through the first openings 33a1 and 33a2 of the lower insulation layer 33. The first pad metal layer 35a may include an outer contact portion 35a1 contacting the first conductivity type semiconductor layer 23 along the periphery of the mesa M and an inner contact portion 35a2 contacting the first conductivity type semiconductor layer 23 in the indent portion. The outer contact portion 35a1 contacts the first conductivity type semiconductor layer 23 near an edge of the substrate 21 along the periphery of the mesa M, and the inner contact portion 35a2 contacts the first conductivity type semiconductor layer 23 in a region surrounded by the outer contact portion 35a1. The outer contact portion 35a1 and the inner contact portion 35a2 may be connected to each other, but the inventive concepts are not limited thereto, and they may be separated from each other.

Meanwhile, the second pad metal layer 35b is disposed on the upper region of the mesa M on the lower insulation layer 33, and is electrically connected to the ohmic reflection layer 31 through the second opening 33b of the lower insulation layer 33. The second pad metal layer 35b may be surrounded by the first pad metal layer 35a, and a boundary region 35ab may be formed therebetween. The lower insulation layer 33 is exposed to the boundary region 35ab, and this boundary region 35ab is covered with the upper insulation layer 37, which will be described later.

The first pad metal layer 35a and the second pad metal layer 35b may be formed of the same material by the same process. The first and second pad metal layers 35a and 35b may include a highly ohmic reflective layer such as an Al layer, and the highly ohmic reflective layer may be formed on an adhesive layer such as Ti, Cr, or Ni. Furthermore, a protective layer of a single layer or a multiple layer structure such as Ni, Cr, Au, or the like may be formed on the highly ohmic reflective layer. The first and second pad metal layers 35a and 35b may have multilayer structures of Cr/Al/Ni/Ti/Ni/Ti/Au/Ti, for example.

The upper insulation layer 37 covers the first and second pad metal layers 35a and 35b. In addition, the upper insulation layer 37 may cover the first conductivity type semiconductor layer 23 along the periphery of the mesa M. However, the upper insulation layer 37 may expose the first conductivity type semiconductor layer 23 along the edge of the substrate 21.

Meanwhile, the upper insulation layer 37 has a first opening 37a exposing the first pad metal layer 35a and a second opening 37b exposing the second pad metal layer 35b. The first opening 37a and the second opening 37b may be disposed in the upper region of the mesa M and may be arranged to face each other. In particular, the first opening 37a and the second opening 37b may be disposed close to both side edges of the mesa M.

The second opening 33b of the lower insulation layer 33 described above may be interposed between the first opening 37a and the second opening 37b of the upper insulation layer 37. The second opening 33b of the lower insulation layer 33

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is separated from both the second opening 37b and the first opening 37a of the upper insulation layer 37.

As shown in FIGS. 1 and 2, a shortest distance L1 from the second opening 37b of the upper insulation layer 37 to the second opening 33b of the lower insulation layer 33 is greater than a shortest distance L2 from the second opening 33b of the lower insulation layer 33 to an edge of the second pad metal layer 35b. Accordingly, the second opening 33b of the lower insulation layer 33 may be relatively farther away from the second opening 37b of the upper insulation layer 37, and thus contamination of the ohmic reflection layer 31 by solder may be prevented more effectively.

The upper insulation layer 37 may be formed of a single layer of SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, without being limited thereto. For example, the upper insulation layer 37 may have a multilayer structure which includes a silicon oxide layer and a silicon nitride layer, and may include a distributed Bragg reflector in which silicon oxide layers and titanium oxide layers are alternately stacked one above another.

The first and second pad metal layers 35a and 35b exposed to the first and second openings 37a and 37b of the upper insulation layer 37 may serve as bonding pads to which the solder is directly bonded. Accordingly, when the light emitting diode according to the present embodiment is bonded to a printed circuit board or a submount using solder, the solder may directly contact the first and second pad metal layers 35a and 35b. In another embodiment, first and second bump pads may be added on the first and second pad metal layers 35a and 35b, which will be described with reference to FIGS. 3 and 4.

FIG. 3 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts, and FIG. 4 is a cross-sectional view taken along the line A-A of FIG. 3.

Referring to FIGS. 3 and 4, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. 1 and 2 except that the light emitting diode further comprises first and second bump pads 39a and 39b.

The first bump pad 39a covers the first pad metal layer 35a exposed through the first opening 37a of the upper insulation layer 37, and the second bump pad 39b covers the second pad metal layer 35b exposed through the second opening 37b of the upper insulation layer 37. Each of the first and second bump pads 39a and 39b may cover the first and second openings 37a and 37b of the upper insulation layer 37. Therefore, the first and second bump pads 39a and 39b may be larger than the first and second openings 37a and 37b.

Further, the second bump pad 39b may cover the upper insulation layer 37 on an upper portion of the second opening 33b of the lower insulation layer 33. As shown in FIG. 3, the second bump pad 39b may have a shape substantially similar to that of the second pad metal layer 35b, and may also have a similar size.

By forming the first and second bump pads 39a and 39b, a diffusion path of the solder is increased, thereby further improving a reliability of the light emitting diode. In particular, in a case that a defect such as a pinhole occurs in an upper insulation layer 37, the second bump pads 39a and 39b prevent the solder from diffusing into the pinhole, thereby improving the reliability of the light emitting diode.

FIG. 5 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. 5, the light emitting diode according to the present embodiment is substantially similar to the light

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emitting diode described with reference to FIGS. 1 and 2 except that indent portions formed inside the mesa M are added and shapes of the second openings 33b of the lower insulation layer 33 is different.

Although two indent portions are shown in the present embodiment, the number of indent portions may be larger. The lower insulation layer 33 has the first opening 33a2 in the indent portion, and the first pad metal layer 35a is connected to the first conductivity type semiconductor layer 23 through the first opening 33a2. Accordingly, as the number of indent portions increases, the number of inner contact portions 35a2 of the first pad metal layer 35a increases, thereby improving current spreading capability.

Meanwhile, a shape of an end terminal portion of the indent portion, which has a round shape as a width becomes wider at the end terminal portion thereof, differs from a shape of the indent portion of FIG. 1. As the shape of the end terminal portion of the indent is formed as described above, the lower insulation layer 33 may be formed in a similar shape. In particular, in a case that the lower insulation layer 33 includes a distributed Bragg reflector, if a width of the distributed Bragg reflector is not widened at the end terminal portion as shown in FIG. 1, a severe double step is formed in a side wall of the distributed Bragg reflector, and the pad metal layer 35a is likely to be cracked as an inclination angle of the side wall becomes large. Accordingly, the lower insulation layer 33 may be formed to have a gentle inclination angle by forming the shape of the end terminal portion of the indent portion and the shape of the end terminal portion of the first opening 33a2 of the lower insulation layer 33 as in the present embodiment, thereby improving yield of the light emitting diode.

In the above embodiments, the second openings 33b of the lower insulation layer 33 have a shape elongated in a direction perpendicular to the indent portion. However, in the present embodiment, the second openings 33a have a circular shape, that is, a convex shape outward. A pattern having the elongated shape in one direction is not easily etched at the end terminal portion thereof, but the second openings 33b having the circular shape as in the present embodiment may be formed easily.

Also in the present embodiment, a shortest distance between the second openings 33b and the second opening 37b of the upper insulation layer 37 may be greater than a shortest distance between the second openings 33b and an edge of the second pad metal layer 35b.

FIG. 6 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. 6, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIG. 5 except that a shape of the second opening 37b of the upper insulation layer 37 is different. That is, in the present embodiment, the second opening 37b of the upper insulation layer 37 has a concave shape 37br corresponding to the shape of the second openings 33b of the lower insulation layer 33.

The second openings 33b of the lower insulation layer 33 may have the circular shape as described above, and thus have a convex shape toward the second opening 37b of the upper insulation layer. The second opening 37b of the upper insulation layer 37 has the concave shape 37br corresponding to the convex shape of the second openings 33b of the lower insulation layer 33.

Accordingly, a shortest distance between the second openings 33b of the lower insulation layer 33 and the second

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openings 37b of the upper insulation layer 37 may be relatively increased, thereby further increasing the diffusion path of the solder.

FIG. 7 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. 7, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIG. 6 except that a shape of the second openings 33b of the lower insulation layer 33 is different. The second openings 33b may have various shapes and may have various convex shapes toward the second openings 37b of the upper insulation layer 37, in particular.

Meanwhile, in the embodiments described with reference to FIGS. 5 to 7, although the first and second bump pads 39a and 39b are not separately described, the first and second bump pads 39a and 39b may also be added in these embodiments as described above with reference to FIGS. 3 and 4. The second bump pad 39b may also cover the second openings 33b of the lower insulation layer 33.

FIG. 8 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts, and FIG. 9 is a cross-sectional view taken along the line B-B of FIG. 8.

Referring to FIGS. 8 and 9, the light emitting diode comprises a substrate 21, a first light emitting cell C1, a second light emitting cell C2, an ohmic reflection layer 31, a lower insulation layer 133, first and second pad metal layers 135a and 135b, a connection metal layer 135c, and an upper insulation layer 137. The light emitting diode may also comprise a preliminary insulation layer 29, and the preliminary insulation layer 29 is integrated into the lower insulation layer 133. In addition, each of the first and second light emitting cells C1 and C2 comprises a first conductivity type semiconductor layer 23 and a mesa M disposed on the first conductivity type semiconductor layer 23, and each mesa M comprises an active layer 25 and a second conductivity type semiconductor layer 27.

The substrate 21 may be a growth substrate for growing a Gallium nitride based semiconductor layer, for example, a sapphire substrate, a Gallium nitride substrate, a SiC substrate, or the like, and particularly a patterned sapphire substrate. The substrate 21 is preferably an insulating substrate, but is not limited to the insulating substrate. However, in a case that light emitting cells C1 and C2 disposed on the substrate 21 are connected to each other in series, the substrate 21 must be insulated from the light emitting cells. Therefore, the substrate 21 may be insulative, or if the substrate 21 is conductive, an insulation material layer is formed between the light emitting cells C1 and C2 and the substrate 21 for the light emitting cells C1 and C2 to be insulated from the substrate 21. The substrate 21 may have a rectangular outer shape as shown in FIG. 8.

The first and second light emitting cells C1 and C2 are disposed on the substrate 21. The first and second light emitting cells C1 and C2 are separated from each other by an isolation region I exposing the substrate 21. Therefore, semiconductor layers of the first light emitting cell C1 and the second light emitting cell C2 are spaced apart from each other. Each of the first and second light emitting cells C1 and C2 are disposed facing each other and may have a square or rectangular shape, respectively. In particular, the first and second light emitting cells C1 and C2 may have elongated rectangular shapes in a direction facing each other.

Each of the first and second light emitting cells C1 and C2 comprises a first conductivity type semiconductor layer 23,

an active layer **25** and a second conductivity type semiconductor layer **27**. The first conductivity type semiconductor layer **23**, the active layer **25** and the second conductivity type semiconductor layer **27** may be formed of III-V nitride semiconductors, for example, nitride semiconductors such as (Al, Ga, In)N. The first conductivity type semiconductor layer **23**, the active layer **25**, and the second conductivity type semiconductor layer **27** may be grown and formed on the substrate **21** in a chamber using a well known method such as Metal Organic Chemical Vapor Deposition (MOCVD). In addition, the first conductivity type semiconductor layer **23** comprises n-type impurities (for example Si, Ge, and Sn), the second conductivity type semiconductor layer **27** comprises p-type impurities (for example, Mg, Sr, and Ba), and vice versa. In one embodiment, the first conductivity type semiconductor layer **23** may comprise GaN or AlGaIn containing Si as a dopant, and the second conductivity type semiconductor layer **27** may comprise GaN or AlGaIn containing Mg as a dopant. Although each of the first conductivity type semiconductor layer **23** and the second conductivity type semiconductor layer **27** is shown as a single layer in the drawings, these layers may be multi-layered and may also comprise a superlattice layer. The active layer **25** may comprise a single quantum well structure or a multiple quantum well structure, and a composition ratio of the nitride-based semiconductor is adjusted so as to emit a desired wavelength. For example, the active layer **25** may emit blue light or ultraviolet light.

An isolation region **I** separates the light emitting cells **C1** and **C2** from each other. Accordingly, the substrate **21** is exposed to the isolation region **I** through semiconductor layers. The isolation region **I** is formed using photolithography and etching processes, the photoresist pattern having a gentle inclined surface is formed and the semiconductor layers are etched using the photoresist pattern as a mask, thereby forming relatively gently inclined side surfaces in the isolation region **I**.

The light emitting cells **C1** and **C2** face each other with the isolation region **I** interposed therebetween. Side surfaces of the light emitting cells **C1** and **C2** facing each other are defined as inner side surfaces, and side surfaces of the light emitting cells other than the inner side surfaces are defined as outer side surfaces. Therefore, the first conductivity type semiconductor layers **23** in the first and second light emitting cells **C1** and **C2** also comprise inner and outer side surfaces, respectively. For example, the first conductivity type semiconductor layer **23** may comprise one inner side surface and three outer side surfaces.

A mesa **M** is disposed on each of first conductivity type semiconductor layers **23**. The mesa **M** may be located within a region surrounded by the side surfaces of the first conductivity type semiconductor layer **23**, and thus regions near edges adjacent to the outer side surfaces of the first conductivity type semiconductor layer **23** are not covered by the mesa **M**, but are exposed to the outside. However, a side surface of the mesa **M** and a side surface of the first conductivity type semiconductor layer **23** on a sidewall of the isolation region **I** may be continuous with each other.

The mesa **M** comprises a second conductivity type semiconductor layer **27** and an active layer **25**. The active layer **25** is interposed between the first conductivity type semiconductor layer **23** and the second conductivity type semiconductor layer **27**.

Ohmic reflection layers **31** are disposed on the second conductivity type semiconductor layers **27** of the first and second light emitting cells **C1** and **C2**, respectively. The ohmic reflection layers **31** contact the second conductivity

type semiconductor layers **27**. The ohmic reflection layer **31** may be disposed substantially over an entire area of the mesa **M** in an upper region of the mesa **M**. For example, the ohmic reflection layer **31** may cover 80% or more, further, 90% or more of the upper region of the mesa **M**.

The ohmic reflection layer **31** may comprise a reflective metallic layer and thus may reflect light generated in the active layer **25** and traveling to the ohmic reflection layer **31**, toward the substrate **21**. For example, the ohmic reflection layer may include Ag or Al. In addition, the ohmic reflection layer **31** may include an Ni layer to form ohmic contact with the second conductivity type semiconductor layer **27**.

Meanwhile, a preliminary insulation layer **29** may cover the mesa **M** in a periphery of the ohmic reflection layer **31**. The preliminary insulation layer **29** may be formed of SiO<sub>2</sub> using a chemical vapor deposition technique, for example, and may cover a side of the mesa **M** and may further cover a partial region of the first conductivity type semiconductor layer **23**. The preliminary insulation layer **29** may be removed at sides of the isolation region **I** as shown in FIG. **8**.

The lower insulation layer **133** covers the ohmic reflection layer **31** and the preliminary insulation layer **29** while covering the mesas **Ms**. The lower insulation layer **133**, in addition, covers the isolation region **I** and sidewalls of the mesa **M** and covers a portion of the first conductivity type semiconductor layer **23** in a periphery of the mesa **M**. As shown in the enlarged cross-sectional view of FIG. **8**, the lower insulation layer **133** may be formed along a shape of protrusions on the substrate **21** in the isolation region **I** in a case that the substrate **21** is a patterned sapphire substrate.

The lower insulation layer **133** is disposed between first and second pad metal layers **135a** and **135b** and the first and second light emitting cells **C1** and **C2**, and has openings **133a**, **133b**, **133c**, and **133d** to provide a passage for the first and second pad metal layers **135a** and **135b** to be connected to the first conductivity type semiconductor layer **23** or the ohmic reflection layer **31**. For example, the lower insulation layer **133** may include first openings **133a** and **133b** exposing the first conductivity type semiconductor layer **23** of the first and second light emitting cells **C1** and **C2**, and second openings **133c** and **133d** exposing the ohmic reflection layer **31** of the first and second light emitting cells **C1** and **C2**. The first opening **133a** exposes the first conductivity type semiconductor layer **23** along outer surfaces of the mesa **M**, and the first opening **133b** exposes the first conductivity type semiconductor layer **23** along inner surfaces of the mesa **M**. The first opening **133a** and the second opening **133c** may be connected each other as shown in FIG. **8**, but the inventive concepts are not limited thereto, they may be separated from each other.

Meanwhile, the second opening **133c** exposes the ohmic reflection layer **31** on the first light emitting cell **C1**, and the second opening **133d** exposes the ohmic reflection layer **31** on the second light emitting cell **C2**. The second opening **133d** may be formed to be smaller than the second opening **133c**. The second openings **133c** and **133d** may generally have elongated shapes along the isolation region **I**, but the inventive concepts are not limited thereto, they may have various shapes.

The lower insulation layer **133** may be formed of an insulating material such as SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, and may be formed as a single layer or multiple layers. Further, the lower insulation layer **133** may comprise a distributed Bragg reflector formed by repeatedly stacking material layers having different refractive indices, for example, SiO<sub>2</sub>/TiO<sub>2</sub>. When the lower insulation layer **133** comprises the distrib-

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uted Bragg reflector, light incident on a region other than the ohmic reflection layer 31 may be reflected and light extraction efficiency may be improved.

Meanwhile, the first pad metal layer 135a, the second pad metal layer 135b and the connection metal layer 135c are disposed on the lower insulation layer 133.

The first pad metal layer 135a is disposed on the first light emitting cell C1 and forms ohmic contact with the first conductivity type semiconductor layer 23. The first pad metal layer 135a may form ohmic contact with the first conductivity type semiconductor layer 23 in a region between an outer side surface of the first conductivity type semiconductor layer 23 and the mesa M along the periphery of the mesa M, as well illustrated in the FIG. 8. Although the first pad metal layer 135a is illustrated as continuously contacting the first conductivity type semiconductor layer 23 along the periphery of the mesa M in the drawings, the present disclosure is not limited thereto, it may intermittently contact the first conductivity type semiconductor layer 23. That is, the first pad metal layer 135a may contact the first conductivity type semiconductor layer 23 in a plurality of regions separated from each other around the mesa M. Further, the first pad metal layer 135a may cover the upper region and the outer side surfaces of the mesa M.

The second pad metal layer 135b is disposed on the second light emitting cell C2 and is connected to the ohmic reflection layer 31 on the second light emitting cell C2 through the second opening 133d of the lower insulation layer 133. The second pad metal layer 135b is disposed on the mesa M and is insulated from the first conductivity type semiconductor layer 23. For example, the second pad metal layer 135b may be separated from the sides of the mesa M on the second light emitting cell C2. The second pad metal layer 135b may substantially cover an entire area of the mesa M on the second light emitting cell C2, and may cover 50% or more of the upper area of the mesa M, for example.

Meanwhile, the connection metal layer 135c may be electrically connected to the ohmic reflection layer 31 on the first light emitting cell C1 through the second opening 133c of the first light emitting cell C1, and may be electrically connected to the first conductivity type semiconductor layer 23 of the second light emitting cell C2 through the first openings 133a and 133b of the second light emitting cell C2. Accordingly, the first and second light emitting cells C1 and C2 are connected to each other in series through the connection metal layer 135c.

The connection metal layer 135c may contact the first conductivity type semiconductor layer 23 on at least one side along an edge of the second light emitting cell C1. In particular, the connection metal layer 135c may contact the first conductivity type semiconductor layer 23 continuously or intermittently along the periphery of the mesa M. In addition, the connection metal layer 135c may surround the second pad metal layer 135b, and a boundary region 133bc may be formed between the connection metal layer 135c and the second pad metal layer 135b. Meanwhile, a boundary region 133ac is formed between the connection metal layer 135c and the first pad metal layer 135a. These boundary regions 133ac and 133bc are covered with an upper insulation layer 137, which will be described later.

The first and second pad metal layers 135a and 135b and the connection metal layer 135c may be formed of the same material by the same process. For example, the first and second pad metal layers 135a and 135b and the connection metal layer 135c may include a highly ohmic reflective layer such as an Al layer, and the highly ohmic reflective layer may be formed on an adhesive layer such as Ti, Cr, or Ni.

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Furthermore, a protective layer of a single layer or a multiple layer structure such as Ni, Cr, Au, or the like may be formed on the highly ohmic reflective layer. The first and second pad metal layers 135a and 135b and the connection metal layer 135c may have multilayer structures of Cr/Al/Ni/Ti/Ni/Ti/Au/Ti, for example.

The upper insulation layer 137 is disposed on the first pad metal layer 135a, the second pad metal layer 135b and the connection metal layer 135c, and has a first opening 137a exposing the first pad metal layer 135a and a second opening 137b exposing the second pad metal layer 135b. The upper insulation layer 137 also covers the first pad metal layer 135a and the connection metal layer 135c connected to the first conductivity type semiconductor layer 23 in the periphery of the mesa M. As shown in FIG. 8, a region between the first pad metal layer and the second connection metal layer 135c and an edge of the first conductivity type semiconductor layer 23 is covered with the upper insulation layer 137. In addition, the upper insulation layer 37 may cover the connection metal layer 135c on the isolation region I, and may be formed to have irregularities according to a shape of the connection metal layer 135c. The upper insulation layer 137 protects the first and second pad metal layers 135a and 135b and the connection metal layer 135c from external environments such as moisture.

Meanwhile, the first opening 137a is formed within an upper region of the first pad metal layer 135a, and thus is separated from the connection metal layer 135c and the second opening 133c of the lower insulation layer 133. In addition, the second opening 137b is also disposed limitedly on the second pad metal layer 135b, and is separated from the connection metal layer 135c. Further, the second opening 137b is separated from the second opening 133d of the lower insulation layer 133. In particular, a shortest distance L1 from the second opening 133d of the lower insulation layer 133 to the second opening 137b of the upper insulation layer 137 is greater than a shortest distance L2 from the second opening 133d of the lower insulation layer 133 to an edge of the second pad metal layer 135b. The second opening 137b of the upper insulation layer 137 is further separated from the second opening 133d than the edge of the second pad metal layer 135b is separated from the second opening 133d, thereby effectively preventing diffusion of the solder within a limited design range.

The upper insulation layer 137 may be formed of a single layer of SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, without being limited thereto. For example, the upper insulation layer 137 may have a multiple layer structure comprising a silicon nitride layer and a silicon oxide layer, or may be a distributed Bragg reflector where a silicon oxide layer and a titanium oxide layer are alternately stacked one above another.

In the present embodiment, the first and second pad metal layers 135a and 135b exposed through the first and second openings 137a and 137b of the upper insulation layer 137 may serve as bonding pads to which the solder is directly bonded. Alternatively, as described with reference to FIGS. 3 and 4, the first and second bump pads may cover the first and second pad metal layers 135a and 135b exposed through the first and second openings 137a and 137b of the upper insulation layer 137, respectively. Further, the first and second bump pads may cover and seal the first and second openings 137a and 137b of the upper insulation layer 137, respectively, and the second bump pad may cover an upper region of the second opening 133d of the lower insulation layer 133 on the second light emitting cell C2.

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FIG. 10 is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. 10, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. 8 and 9 except that an overall shape of the light emitting diode is different, and shapes of the openings 133a, 133b, and 133d of the lower insulation layer are different.

First, the first and second light emitting cells C1 and C2 have elongated rectangular shapes perpendicular to a direction where the first and second light emitting cells C1 and C2 are facing each other. Accordingly, the first pad metal layer 135a and the second pad metal layer 135b also have vertically long shapes along the shape of the light emitting cells.

Meanwhile, the lower insulation layer 133 has the first opening 133a exposing the first conductivity type semiconductor layer 23 along the outer side surface of the mesa M on the first light emitting cell C1 in the previous embodiments, but the first opening 133a extends along the inner side surface of the mesa M on the first light emitting cell C1 to further expose the first conductivity type semiconductor layer 23 in the present embodiment. In this case, the end terminal portion of the first opening 133a formed along the inner side surface of the first light emitting cell C1 may have a round shape with a wider width. The first opening 133a forms to expose the first conductivity type semiconductor layer 23 along the inner side surface of the mesa M, so that a contact area of the first pad metal layer 135a may increase, thereby improving current spreading capability.

The first opening 133b exposing the first conductivity type semiconductor layer 23 along the inner surface of the mesa M of the second light emitting cell C2 in the previous embodiment is formed along the isolation region I and is connected to the first opening 133a. In the present embodiment, however, the first opening 133b is separated from the first opening 133a. Further, both end terminal portions of the first opening 133b may have round shapes with a wider width.

Moreover, in the present embodiment, a plurality of second openings 133d of the lower insulation layer 133 may be formed.

FIG. 11 is a schematic plan view illustrating a light emitting diode according to an exemplary embodiment of the inventive concepts, and FIG. 12 is a cross-sectional view taken along the line C-C of FIG. 11.

Referring to FIGS. 11 and 12, the light emitting diode comprises a substrate 21, a first conductivity type semiconductor layer 23, an active layer 25, a second conductivity type semiconductor layer 27, an ohmic reflection layer 31, a lower insulation layer 33, a first pad metal layer 35a, a second pad metal layer 35b, and an upper insulation layer 37. Furthermore, the light emitting diode may further comprise a preliminary insulation layer 29, and may further comprise a first bump pad 39a and a second bump pad 39b.

The substrate 21 may be selected from any substrates suitable for growth of gallium nitride semiconductor layers thereon. The substrate 21 may include, for example, a sapphire substrate, a gallium nitride substrate, a SiC substrate, and the like, and may be a patterned sapphire substrate. The substrate 21 may have a rectangular or square shape, as shown in the plan view of FIG. 1, without being limited thereto. The size of the substrate 21 is not particularly limited and may be selected in various ways.

The first conductivity type semiconductor layer 23 is disposed on the substrate 21. The first conductivity type semiconductor layer 23 may be a gallium nitride semicon-

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ductor layer grown on the substrate 21. The first conductivity type semiconductor layer 23 may be a gallium nitride semiconductor layer doped with an impurity, for example, Si.

A mesa M is disposed on each of first conductivity type semiconductor layers 23. The mesa M may be located within a region surrounded by the side surfaces of the first conductivity type semiconductor layer 23, and thus regions near edges of the first conductivity type semiconductor layer 23 are not covered by the mesa M, but are exposed to the outside.

The mesa M comprises a second conductivity type semiconductor layer 27 and an active layer 25. The active layer 25 is interposed between the first conductivity type semiconductor layer 23 and the second conductivity type semiconductor layer 27. The active layer 25 may have a single quantum well structure or a multiple quantum well structure. A composition and a thickness of the well layer in the active layer 25 determine a wavelength of generated light. In particular, by controlling the composition of the well layer, it is possible to provide an active layer generating ultraviolet light, blue light or green light.

Meanwhile, the second conductivity type semiconductor layer 27 may be a gallium nitride semiconductor layer doped with p-type dopants, for example, Mg. Each of the first conductivity type semiconductor layer 23 and the second conductivity type semiconductor layer 27 may be composed of a single layer or multiple layers and may include superlattice layers. The first conductivity type semiconductor layer 23, the active layer 25 and the second conductivity type semiconductor layer 27 may be grown on the substrate in a chamber by a well-known method, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE).

As shown in FIG. 11, the mesa M may have an indent portion 30 intruding therein, and an upper surface of the first conductivity type semiconductor layer 23 may be exposed by the indent portion 30. The indent portion 30 may be formed long from one side edge of the mesa M toward the other side opposite to the mesa M. A length of the indent portion 30 is not particularly limited, and may be 1/2 of the length of the mesa M or longer than that. Also, although two indent portions 30 are shown in FIG. 11, the number of indent portion 30 may be one or three or more. Accordingly, as the number of indent portions 30 increases, the number of inner contact portions 35a2 of the first pad metal layer 35a, which will be described later, increases, thereby improving current spreading capability.

Meanwhile, the indent portion 30 has a round shape as a width becomes wider at an end terminal portion thereof. As the shape of the end terminal portion of the indent is formed as described above, the lower insulation layer 33 may be formed in a similar shape. In particular, in a case that the lower insulation layer 33 includes a distributed Bragg reflector, if a width of the distributed Bragg reflector is not widened at the end terminal portion as shown in FIG. 11, a severe double step is formed in a side wall of the distributed Bragg reflector, and the pad metal layer 35a is likely to be cracked as an inclination angle of the side wall becomes large. Accordingly, the lower insulation layer 33 may be formed to have a gentle inclination angle by forming the shape of the end terminal portion of the indent portion 30 and the shape of the end terminal portion of the first opening 33a2 of the lower insulation layer 33 as in the present embodiment, thereby improving yield of the light emitting diode.



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The ohmic reflection layer 31 is disposed over the mesa M to contact the second conductivity type semiconductor layer 27. The ohmic reflection layer 31 may be disposed substantially over an entire area of the mesa M in an upper region of the mesa M. For example, the ohmic reflection layer 31 may cover 80% or more, further, 90% or more of the upper region of the mesa M.

The ohmic reflection layer 31 may include a metal layer having reflectivity and thus can reflect light generated from the active layer 25 and travelling to the ohmic reflection layer 31, toward the substrate 21. For example, the ohmic reflection layer 31 may be composed of a single reflection metal layer, but is not limited thereto. Alternatively, the ohmic reflection layer 31 may include an ohmic layer and a reflection layer. The ohmic layer may be a metal layer such as a Ni layer or a transparent oxide layer such as an ITO layer, and the reflection layer may be a metal layer having high reflectance such as an Ag or Al layer.

Meanwhile, a preliminary insulation layer 29 may cover the mesa M in a periphery of the ohmic reflection layer 31. The preliminary insulation layer 29 may be formed of SiO<sub>2</sub>, and may cover a side of the mesa M and may further cover a partial region of the first conductivity type semiconductor layer 23. In another embodiment, the preliminary insulation layer 29 may be disposed only on the periphery of the ohmic reflection layer 31 in the upper region of the mesa M.

The lower insulation layer 33 covers the mesa M and the ohmic reflection layer 31. The lower insulation layer 33 may also cover the first conductivity type semiconductor layer 23 along a periphery of the mesa M, and may cover the first conductivity type semiconductor layer 23 in the indent portion 30 inside the mesa M. The lower insulation layer 33 covers side surfaces of the mesa M in particular.

Meanwhile, the lower insulation layer 33 has first and second openings 33a1 and 33a2 exposing the first conductivity type semiconductor layer 23 and a second opening 33b exposing the ohmic reflection layer 31. The first opening 33a1 exposes the first conductivity type semiconductor layer 23 along the periphery of the mesa M, and the first opening 33a2 exposes the first conductivity type semiconductor layer 23 in the indent portion 30. As shown in FIG. 11, the first opening 33a1 and the first opening 33a2 may be connected to each other. However, the inventive concepts are not limited thereto, but the first opening 33a1 and the first opening 33a2 may be separated from each other.

The second opening 33b exposes the ohmic reflection layer 31. A plurality of second openings 33b may be formed, and these second openings 33b may be disposed near one edge of the substrate 21 so as to face the indent portion 30. Locations of the second openings 33b will be described later.

Meanwhile, the lower insulation layer 33 covers the preliminary insulation layer 29 and is integrated with the preliminary insulation layer 29. Unless specifically stated, it should be understood that the preliminary insulation layer 29 is included in the lower insulation layer 33. The lower insulation layer 33 may be formed of a single layer of SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, without being limited thereto. For example, the lower insulation layer 33 may have a multilayer structure which includes a silicon oxide layer and a silicon nitride layer, and may include a distributed Bragg reflector in which silicon oxide layers and titanium oxide layers are alternately stacked one above another.

The first pad metal layer 35a is disposed on the lower insulation layer 33 and is insulated from the mesa M and the ohmic reflection layer 31 by the lower insulation layer 33. The first pad metal layer 35a contacts the first conductivity type semiconductor layer 23 through the first openings 33a

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and 33a2 of the lower insulation layer 33. The first pad metal layer 35a may include an outer contact portion 35a1 contacting the first conductivity type semiconductor layer 23 along the periphery of the mesa M and an inner contact portion 35a2 contacting the first conductivity type semiconductor layer 23 in the indent portion 30. The outer contact portion 35a1 contacts the first conductivity type semiconductor layer 23 near an edge of the substrate 21 along the periphery of the mesa M, and the inner contact portion 35a2 contacts the first conductivity type semiconductor layer 23 in a region surrounded by the outer contact portion 35a1. The outer contact portion 35a1 and the inner contact portion 35a2 may be connected to each other, but the inventive concepts are not limited thereto, and they may be separated from each other.

Meanwhile, the second pad metal layer 35b is disposed on the upper region of the mesa M on the lower insulation layer 33, and is electrically connected to the ohmic reflection layer 31 through the second opening 33b of the lower insulation layer 33. The second pad metal layer 35b may be surrounded by the first pad metal layer 35a, and a boundary region 35ab may be formed therebetween. The lower insulation layer 33 is exposed to the boundary region 35ab, and this boundary region 35ab is covered with the upper insulation layer 37, which will be described later.

The first pad metal layer 35a and the second pad metal layer 35b may be formed of the same material by the same process. The first and second pad metal layers 35a and 35b may include an ohmic reflection layer such as an Al layer, and the ohmic reflection layer may be formed on an adhesive layer such as Ti, Cr, or Ni. Furthermore, a protective layer of a single layer or a multiple layer structure such as Ni, Cr, Au, or the like may be formed on the ohmic reflection layer. The first and second pad metal layers 35a and 35b may have multilayer structures of Cr/Al/Ni/Ti/Ni/Ti/Au/Ti, for example.

The upper insulation layer 37 covers the first and second pad metal layers 35a and 35b. In addition, the upper insulation layer 37 may cover the first conductivity type semiconductor layer 23 along the periphery of the mesa M. However, the upper insulation layer 37 may expose the first conductivity type semiconductor layer 23 along the edge of the substrate 21.

Meanwhile, the upper insulation layer 37 has a first opening 37a exposing the first pad metal layer 35a and a second opening 37b exposing the second pad metal layer 35b. The first opening 37a and the second opening 37b may be disposed in the upper region of the mesa M and may be arranged to face each other. In particular, the first opening 37a and the second opening 37b may be disposed close to both side edges of the mesa M.

The second opening 33b of the lower insulation layer 33 described above may be disposed near the second opening 37b of the upper insulation layer 37. However, the second opening 33b of the lower insulation layer 33 is separated from the first opening 37a of the upper insulation layer 37 and also from the second opening 37b of the upper insulation layer 37 not to overlap with each other. Accordingly, even if a solder intrudes through the second opening 37b of the upper insulation layer 37, it is possible to prevent the solder from diffusing into the second opening 33b of the lower insulation layer 33, thereby preventing the contamination of the ohmic reflection layer 31 by the solder.

The upper insulation layer 37 may be formed of a single layer of SiO<sub>2</sub> or Si<sub>3</sub>N<sub>4</sub>, without being limited thereto. For example, the upper insulation layer 37 may have a multilayer structure which includes a silicon oxide layer and a



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silicon nitride layer, and may include a distributed Bragg reflector in which silicon oxide layers and titanium oxide layers are alternately stacked one above another.

Meanwhile, the first bump pad **39a** electrically contacts the first pad metal layer **35a** exposed through the first opening **37a** of the upper insulation layer **37**, and the second bump pad **39b** electrically contacts the second pad metal layer **35b** exposed through the second opening **37b**. As shown in FIG. **11**, the first bump pad **39a** covers and seals all of the first openings **37a** of the upper insulation layer **37**, and the second bump pad **39b** covers and seals all of the second openings **37b** of the upper insulation layer **37**. Further, the second bump pad **39b** covers an upper region of the second opening **33b** of the lower insulation layer **33**. That is, the second bump pad **39b** covers the upper insulation layer **37** in the upper region of the second opening **33b** of the lower insulation layer **33**. The second bump pad **39b** may cover all of the second openings **33b** of the lower insulation layer **33**, but the inventive concepts are not limited thereto, and a portion of the openings **33b** may be disposed outside of the second bump pad **39b**.

Further, as shown in FIG. **11**, the second bump pad **39b** may be disposed within an upper region of the second pad metal layer **35b**, but the inventive concepts are not limited thereto, a portion of the second bump pad **39b** may overlap with the first pad metal layer **35a**. However, the upper insulation layer **37** may be disposed between the first pad metal layer **35a** and the second bump pad **39b** to insulate them.

FIGS. **13A** and **13B** are schematic plan views illustrating the second openings **33b** and **37b** of the lower insulation layer and the upper insulation layer in relation to the embodiment of FIG. **11**.

FIG. **13A** shows the arrangement of second openings **33b** of the lower insulation layer **33** and second openings **37b** of the upper insulation layer **37** arranged in the same manner as the embodiment of FIG. **11**, and FIG. **13B** shows another modification.

Referring to FIGS. **13A** and **13B**, L1 represents a shortest distance between the second opening **33b** and the second opening **37b**, L2 represents a shortest distance between the second openings **33b**, and L3 represents a shortest distance between the second openings **37b**.

In FIG. **13A**, the second openings **33b** and the second openings **37b** are alternately disposed at an upper side and a lower side, and the second openings **33b** and the second openings **37b** are alternately disposed in a lateral direction. In this arrangement, L1 is relatively short compared to L2 or L3. At least one of the second openings **37b** is disposed between the second openings **33b**.

In FIG. **13B**, the same second openings **33b** are disposed at an upper side and the same second openings **37b** are disposed at a lower side, or vice versa, and the second openings **33b** and the second openings **37b** are alternately disposed in the lateral direction. In this arrangement, L1 is relatively long compared to L2 or L3. Also, at least one of the second openings **37b** is disposed between the second openings **33b**.

Considering the diffusion of the solder, it is advantageous to dispose the second opening **33b** away from the second opening **37b**. Therefore, the arrangement of FIG. **13B** may be more advantageous than the arrangement of FIG. **13A**.

In the drawings of FIGS. **13A** and **13B**, it is illustrated that the distance between the openings disposed at an upper side and a lower side is shorter than the distance between the openings disposed in the lateral direction. Alternatively, the distance between the openings disposed in the lateral direc-

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tion may be shorter. In this case, in order to dispose the second opening **33b** and the second opening **37b** relatively farther apart, the second openings **33a** and **37b** are disposed vertically, the second openings **33b** are continuously disposed laterally, and the second openings **37b** are continuously disposed laterally, and thus it is more advantageous to prevent diffusion of the solder.

However, the number and the arrangement of the second openings **33b** and the second openings **37b** may be selected in consideration of an efficiency of current spreading and a symmetry of light emission pattern as well as contamination of the ohmic reflection layer **31** by the solder, and thus may variously modified.

FIG. **14** is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. **14**, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. **11** and **1** except that shapes of the first pad metal layer **35a** and the second pad metal layer **35b**, an arrangement of the second opening **33b** of the lower insulation layer **33**, and a shape of the second bump pad **39b** are different.

That is, the second openings **33b** of the lower insulation layer **33** are added near the indent portion **30** in comparison with the embodiment of FIG. **11**. The added second openings **33b** are disposed closer to a center portion of the mesa M than an edge of the mesa M in which the second openings **37b** of the upper insulation layer **37** are disposed. Although it is illustrated that the added second openings **33b** are located apart from the center portion of the mesa M toward the edge side at which the second openings **37b** of the upper insulation layer **37** are disposed. However, the inventive concepts are not limited thereto, the added second openings **33b** may be disposed at the center portion of the mesa M or may be disposed apart from the center portion toward an edge side of the mesa M where the first bump pad **39a** is disposed.

The second pad metal layer **35b** is disposed to cover the second openings **33b**. As shown in FIG. **14**, the second pad metal layer **35b** may have a concavoconvex shape having a concave portion near the indent portion **30**. The first pad metal layer **35a** may have a concavoconvex shape so that the boundary region **35ab** between the first pad metal layer **35a** and the second pad metal layer **35b** has a constant width.

Meanwhile, the second bump pad **39b** may have a shape similar to that of the second pad metal layer **35b**, and may cover upper regions of the added second openings **33b**.

According to the present embodiment, the second openings **33b** of the lower insulation layer **33** may be distributed relatively widely, thereby helping to spread the current in the ohmic reflection layer **31**.

FIG. **15** is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. **15**, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIG. **14** except that arrangements of the second openings **33b** and the second openings **37b** and shapes of the first and second bump pads **39a** and **39b** are different.

That is, the first bump pad **39a** extends to an upper region of the second pad metal layer **35b**, and covers at least a portion of the upper regions of the second openings **33b**. However, the second openings **37b** are separated from the first bump pads **39a** and disposed outside of the first bump

pads **39a**. The second bump pad **39b** may cover the second openings **37b** and may also cover a portion of the second openings **33b**.

In the present embodiment, the first bump pad **39a** partially overlaps with the second pad metal layer **35b**. Alternatively, the second bump pad **39b** may partially overlap with the first pad metal layer **35a**.

FIG. **16** is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts, and FIG. **17** is a cross-sectional view taken along the line D-D of FIG. **16**.

Referring to FIGS. **16** and **17**, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. **11** and **12** except that a via hole is formed instead of the indent portion **30**.

That is, the mesa **M** has a via hole **230** exposing the first conductivity type semiconductor layer **23** through the second conductivity type semiconductor layer **27** and the active layer **25**. The via hole **230** is surrounded by the second conductivity type semiconductor layer **27** and the active layer **25**. The mesa **M** may have a plurality of via holes **230** and the plurality of via holes **230** may be regularly arranged inside the mesa **M**. In FIG. **16**, nine via holes **230** are shown, but the inventive concepts are not particularly limited thereto. Meanwhile, as shown in FIG. **16**, concave portions may be formed in the mesa **M** along the edge of the mesa **M**.

The lower insulation layer **133** may cover the mesa **M** and the ohmic reflection layer **31**, and may comprise a first opening **233a1** exposing the first conductivity type semiconductor layer **23** in an outside of the mesa **M** and a first opening **233a2** exposing the first conductivity type semiconductor layer **23** in the via hole **230**. In addition, the lower insulation layer **233** has second openings **233b** exposing the ohmic reflection layer **31** on the mesa **M**.

A first pad metal layer **235a** comprises outer contact portions **235a1** contacting the first conductivity type semiconductor layer **23** through the first openings **233a1** of the lower insulation layer **233** and inner contact portions **235a2** contacting the first conductivity type semiconductor layer **23** through the first openings **233a2**. The inner contact portions **235a2** may be evenly distributed in the mesa **M** for current spreading. Meanwhile, as shown in FIG. **16**, the outer contact portions **235a1** may be distributed on three sides except one side of the mesa **M**, but the present disclosure is not limited thereto, the outer contact portions **235a1** may be distributed on four sides. Further, the outer contact portion **235a1** may be continuously formed along the mesa **M** as in the previous embodiments.

Meanwhile, a second pad metal layer **235b** is electrically connected to the ohmic reflection layer **31** through the second openings **233b** of the lower insulation layer **233**. The second pad metal layer **235b** is separated from the via holes **230**. As shown in FIG. **16**, the second openings **233b** of the lower insulation layer **233** may be disposed in a region surrounded by the via holes **230**, and the second pad metal layer **235b** may cover the second openings **233b**. Accordingly, the first pad metal layer **235a** and the second pad metal layer **235b** may be formed to have concavoconvex shapes.

The upper insulation layer **37** has a first opening **37a** exposing the first pad metal layer **235a** and a second opening **37b** exposing the second pad metal layer **235b**. The first openings **37a** and the second openings **37b** may be disposed near edges facing each other. In particular, the second openings **37b** of the upper insulation layer **37** are separated from the second openings **233b** of the lower insulation layer **233** not to overlap each other.

Meanwhile, the first bump pad **239a** and the second bump pad **239b** are electrically connected to the first pad metal layer **235a** and the second pad metal layer **235b** through the first and second openings **37a** and **37b** of the upper insulation layer **37**, respectively.

As shown in FIG. **16**, the second bump pad **239b** may have a concavoconvex shape so as to surround the via holes **230**. In addition, the second bump pad **239b** may cover at least a portion of the second openings **233b** of the lower insulation layer **233**. As shown in FIG. **17**, for example, the second bump pad **239b** may cover an upper region of the second openings **233b** of the lower insulation layer **233** except the second openings **233b** disposed near the center portion of the mesa **M**. However, the inventive concepts are not limited thereto, the second bump pad **239b** may cover the entire upper region of the second openings **233b**.

FIG. **18** is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts, and FIG. **19** is a cross-sectional view taken along the line E-E of FIG. **18**.

Referring to FIGS. **18** and **19**, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. **11** and **12** except that the light emitting diode includes a plurality of mesas **M1**, **M2**, and **M3**.

That is, the light emitting diode in the embodiments in FIGS. **11** and **12** has a structure in which a single mesa **M** includes the indent portion **30** whereas the light emitting diode according to the present embodiment has a structure in which the mesas **M1**, **M2**, and **M3** are disposed to be separated from each other. The mesas **M1**, **M2**, and **M3** may be disposed to be flush with each other along a longitudinal direction. This structure corresponds to, for example, a structure in which the indent portion **30** is connected from one side edge to the other edge side in the embodiment of FIG. **11**. As a result, shapes of the ohmic reflection layer **331** and the first and second pad metal layers **335a** and **335b** are modified.

The ohmic reflection layer **331** is located on each of the mesas **M1**, **M2**, and **M3** and forms ohmic contact with each of the second conductivity type semiconductor layers **27**. Each of the ohmic reflection layers **331** is disposed within upper regions of the mesas **M1**, **M2**, and **M3**, respectively, and is separated from each other.

Meanwhile, a preliminary insulation layer **329** may cover the mesas **M1**, **M2**, and **M3** in a periphery of the ohmic reflection layer **331**. The preliminary insulation layer **329** may be formed of  $\text{SiO}_2$ , and may cover sides of the mesas **M1**, **M2**, and **M3**, and may further cover a partial region of the first conductivity type semiconductor layer **23**. In another embodiment, the preliminary insulation layer **329** may be disposed only on the periphery of the ohmic reflection layer **331** in the upper regions of the mesas **M1**, **M2**, and **M3**.

The lower insulation layer **333** covers the mesas **M**, **M2**, and **M3** and the ohmic reflection layers **331**. The lower insulation layer **333** has first openings **333a1** exposing the first conductivity type semiconductor layer **23** along an edge of the substrate **21** and first openings **333a2** exposing the first conductivity type semiconductor layer **23** between the mesas **M1**, **M2**, and **M3**.

In addition, the lower insulation layer **333** has second openings **333b** exposing the ohmic reflection layer **331** on each of the mesas **M1**, **M2**, and **M3**. A shape of the second openings **333b** may be circular in the same manner as in the embodiment of FIG. **11**, but is not limited thereto, the shape

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of the second openings **333b** may be a long rounded rectangular shape as shown in FIG. **18**.

The first pad metal layer **335a** covers the mesas **M1**, **M2**, and **M3** and is electrically connected to the first conductivity type semiconductor layer **23** through the first openings **333a1** and **333a2**. The first pad metal layer **335a** may have an outer contact portion **335a1** to contact the first conductivity type semiconductor layer **23** through the first opening **333a1** and inner contact portions **335a2** between the mesas **M1**, **M2**, and **M3** to contact the first conductivity type semiconductor layer **23** through the first openings **333a2**.

Meanwhile, second pad metal layers **335b** are disposed on the mesas **M1**, **M2**, and **M3**, respectively. Each of the second pad metal layers **335b** may be surrounded by the first pad metal layer **335a**, and thus a ring shaped boundary region **335ab** may be formed on each of the mesas **M1**, **M2**, and **M3**.

The upper insulation layer **337** covers the first pad metal layer **335a** and the second pad metal layer **335b**, and has first openings **337a** exposing the first pad metal layer **335a** and second openings **337b** exposing the second pad metal layer **335b**. The second pad metal layers **335b** on each of the mesas **M1**, **M2**, and **M3** are exposed through the second openings **337b**. The first openings **337a** may also be disposed on each of the mesas **M1**, **M2**, and **M3**.

Meanwhile, a first bump pad **339a** and a second bump pad **339b** may be formed over the mesas **M1**, **M2**, and **M3**, respectively. The first bump pad **339a** contacts the first pad metal layer **335a** through the first openings **337a** of the upper insulation layer **337**, and the second bump pad **339b** contacts the second pad metal layer **335b** through the second openings **337b** of the upper insulation layer **337**. The second bump pad **339b** may also cover an upper region of the second openings **333b** of the lower insulation layer **333**.

In the present embodiment, the second openings **333b** of the lower insulation layer **333** and the second openings **337b** of the upper insulation layer **337** may be arranged in various ways. In the present embodiment, the second openings **333b** and **337b** are arranged in the same number on the mesas **M1**, **M2**, and **M3**, and the second openings **333b** and **337b** are arranged over and under, and have a mirror symmetrical structure as a whole.

FIGS. **20A** and **20B** are schematic plan views illustrating various arrangements of the second openings **333b** of the lower insulation layer **333** and the second openings **337b** of the upper insulation layer **337**.

Referring to FIG. **20A**, the second openings **333b** of the lower insulation layer **333** and the second openings **337b** of the upper insulation layer **337** are disposed with their locations shifted from each other in comparison with the embodiment of FIG. **18**. These openings **333b** and **337b** are also disposed to have the mirror symmetrical structure as a whole.

Referring to FIG. **20B**, the second openings **333b** of the lower insulation layer **333** are disposed over and under each other on the first mesa **M1** and the third mesa **M3**, and the second openings **337b** of the upper insulation layer **337** are also disposed over and under. Meanwhile, the second opening **333b** and the second opening **337b** are disposed over and under on the second mesa **M2**. In the embodiment of FIG. **20B**, these openings **333b** and **337b** are also disposed to have the mirror symmetrical structure as a whole.

The second openings **333b** and **337b** disposed to have the mirror symmetrical structure, so that a symmetrical light emission pattern may be realized. However, the inventive concepts are not limited thereto, and arrangements of the

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openings **333b** and **337b** may be variously modified, and may not have the mirror symmetrical structure.

FIG. **21** is a schematic plan view illustrating a light emitting diode according to another embodiment of the inventive concepts.

Referring to FIG. **21**, the light emitting diode according to the present embodiment is substantially similar to the light emitting diode described with reference to FIGS. **18** and **19** except that the number of mesas and a shape of the first opening **437a** of the upper insulation layer are different. To avoid the repeated contents described with reference to FIGS. **18** and **19**, the detailed description thereof will be omitted, and the difference will be explained mainly.

In the present embodiment, four mesas **M1** to **M4** are disposed on the first conductivity type semiconductor layer **23**. However, the number of mesas is not particularly limited, and may be more or less.

Meanwhile, the first openings **437a** of the upper insulation layer may be circular as shown in FIG. **18**, or may have a long rounded rectangular shape as shown in FIG. **21**. The second opening **337b** of the upper insulation layer may have the same shape, but sizes of the first opening **437a** and the second opening **337b** may be different from each other or may be the same.

By forming the shape of the first opening **437a** with the long rounded rectangular shape, it is possible to secure a sufficient distance between the first openings **437a** disposed laterally and adjacent to each other while increasing the size of the opening. Furthermore, the shape of the first opening **437a** is not particularly limited thereto, but the shape may be variously modified.

Meanwhile, the second opening **337b** of the upper insulation layer and the second opening **333b** of the lower insulation layer are disposed vertically and laterally each other in the present embodiment. Unlike the embodiment of FIG. **18**, the second opening **337b** of the upper insulation layer and the second opening **333b** of the lower insulation layer on each of the mesas **M1** to **M4** may be disposed in the same pattern in the present embodiment. Accordingly, an arrangement of the openings **333b** and **337b** is not mirror symmetrical. The arrangement of the openings **333b** and **337b** may be variously modified as described above with reference to FIGS. **20A** and **20B**. In addition, a shortest distance between the second opening **333b** of the lower insulation layer and the second opening **337b** of the upper insulation layer may be arranged to be greater than a shortest distance between the second openings **333b** of the lower insulation layer or a shortest distance of the second openings **337b** of the upper insulating layer.

A first bump pad **439a** is formed over the mesas **M1** to **M4** and covers the first openings **437a** of the upper insulation layer. The first bump pad **439a** is connected to the first pad metal layer **335a** as in the embodiment of FIG. **18**. Meanwhile, a cathode mark **Mc** for displaying a cathode on the first bump pad **439a** may be formed. For example, as shown in FIG. **21**, a corner portion may be omitted from the first bump pad **439a** having a rectangular shape to form the cathode mark **Mc**.

A second bump pad **439b** is formed over the mesas **M1** to **M4** and covers the second openings **337b** of the upper insulation layer. The second bump pad **439b** is connected to the second pad metal layers **335b** as in the embodiment of FIG. **18**. The second bump pad **439b** may also cover the upper region of the second openings **333b** of the lower insulation layer.

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FIG. 22 is an exploded perspective view illustrating a lighting apparatus to which a light emitting diode according to an exemplary embodiment of the inventive concepts is applied.

Referring to FIG. 22, the lighting apparatus according to the present embodiment includes a diffusive cover 1010, a light emitting diode module 1020, and a body 1030. The body 1030 may receive the light emitting diode module 1020 and the diffusive cover 1010 may be disposed on the body 1030 to cover an upper surface of the light emitting diode module 1020.

The body 1030 may have any shape so long as the body can supply electric power to the light emitting diode module 1020 while receiving and supporting the light emitting diode module 1020. For example, as shown in the drawing, the body 1030 may include a body case 1031, a power supply 1033, a power supply case 1035, and a power source connection 1037.

The power supply 1033 is received in the power supply case 1035 to be electrically connected to the light emitting diode module 1020, and may include at least one IC chip. The IC chip may regulate, change or control electric power supplied to the light emitting diode module 1020. The power supply case 1035 may receive and support the power supply 1033, and the power supply case 1035 having the power supply 1033 secured therein may be disposed within the body case 1031. The power source connection 1037 is disposed at a lower end of the power supply case 1035 and is coupled thereto. Accordingly, the power source connection 1037 is electrically connected to the power supply 1033 within the power supply case 1035 and can serve as a passage through which power can be supplied from an external power source to the power supply 1033.

The light emitting diode module 1020 includes a substrate 1023 and a light emitting diode 1021 disposed on the substrate 1023. The light emitting diode module 1020 may be disposed at an upper portion of the body case 1031 and electrically connected to the power supply 1033.

As the substrate 1023, any substrate capable of supporting the light emitting diode 1021 may be used without limitation. For example, the substrate 1023 may include a printed circuit board having interconnects formed thereon. The substrate 1023 may have a shape corresponding to a securing portion formed at the upper portion of the body case 1031 so as to be stably secured to the body case 1031. The light emitting diode 1021 may include at least one of the light emitting diodes according to the exemplary embodiments described above.

The diffusive cover 1010 is disposed on the light emitting diode 1021 and may be secured to the body case 1031 to cover the light emitting diode 1021. The diffusive cover 1010 may be formed of a light transmitting material and light orientation of the lighting apparatus may be adjusted through regulation of the shape and optical transmissivity of the diffusive cover 1010. Thus, the diffusive cover 1010 may be modified to have various shapes depending on usage and applications of the lighting apparatus.

FIG. 23 is a cross-sectional view illustrating a display apparatus to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

The display apparatus according to the present embodiment includes a display panel 2110, a backlight unit supplying light to the display panel 2110, and a panel guide supporting a lower edge of the display panel 2110.

The display panel 2110 is not particularly limited and may be, for example, a liquid crystal panel including a liquid

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crystal layer. Gate driving PCBs may be further disposed at the periphery of the display panel 2110 to supply driving signals to a gate line. Here, the gate driving PCBs may be formed on a thin film transistor substrate instead of being formed on separate PCBs.

The backlight unit includes a light source module which includes at least one substrate and a plurality of light emitting diodes 2160. The backlight unit may further include a bottom cover 2180, a reflective sheet 2170, a diffusive plate 2131, and optical sheets 2130.

The bottom cover 2180 may be open at an upper surface thereof to receive the substrate, the light emitting diodes 2160, the reflective sheet 2170, the diffusive plate 2131, and the optical sheets 2130. In addition, the bottom cover 2180 may be coupled to the panel guide. The substrate may be disposed under the reflective sheet 2170 to be surrounded by the reflective sheet 2170. Alternatively, when a reflective material is coated on a surface thereof, the substrate may be disposed on the reflective sheet 2170. Further, a plurality of substrates may be arranged parallel to one another, without being limited thereto. However, it should be understood that the backlight unit includes a single substrate.

The light emitting diodes 2160 may include at least one of the light emitting diodes according to the exemplary embodiments described above. The light emitting diodes 2160 may be regularly arranged in a predetermined pattern on the substrate. In addition, a lens 2210 may be disposed on each of the light emitting diodes 2160 to improve uniformity of light emitted from the plurality of light emitting diodes 2160.

The diffusive plate 2131 and the optical sheets 2130 are disposed above the light emitting diode 2160. Light emitted from the light emitting diodes 2160 may be supplied in the form of sheet light to the display panel 2110 through the diffusive plate 2131 and the optical sheets 2130.

In this way, the light emitting diodes according to the exemplary embodiments may be applied to direct type displays like the display apparatus according to the present embodiment.

FIG. 24 is a cross-sectional view illustrating a display apparatus to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

The display apparatus according to the present embodiment includes a display panel 3210 on which an image is displayed, and a backlight unit disposed at a rear side of the display panel 3210 and emitting light thereto. Further, the display apparatus includes a frame 240 supporting the display panel 3210 and receiving the backlight unit, and covers 3240, 3280 surrounding the display panel 3210.

The display panel 3210 is not particularly limited and may be, for example, a liquid crystal panel including a liquid crystal layer. A gate driving PCB may be further disposed at the periphery of the display panel 3210 to supply driving signals to a gate line. Here, the gate driving PCB may be formed on a thin film transistor substrate instead of being formed on a separate PCB. The display panel 3210 is secured by the covers 3240, 3280 disposed at upper and lower sides thereof, and the cover 3280 disposed at the lower side of the display panel 3210 may be coupled to the backlight unit.

The backlight unit supplying light to the display panel 3210 includes a lower cover 3270 partially open at an upper surface thereof, a light source module disposed at one side inside the lower cover 3270, and a light guide plate 3250 disposed parallel to the light source module and converting spot light into sheet light. In addition, the backlight unit

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according to the present embodiment may further include optical sheets **3230** disposed on the light guide plate **3250** to spread and collect light, and a reflective sheet **3260** disposed at a lower side of the light guide plate **3250** and reflecting light traveling in a downward direction of the light guide plate **3250** towards the display panel **3210**.

The light source module includes a substrate **3220** and a plurality of light emitting diodes **3110** arranged at constant intervals on one surface of the substrate **3220**. As the substrate **3220**, any substrate capable of supporting the light emitting diodes **3110** and being electrically connected thereto may be used without limitation. For example, the substrate **3220** may include a printed circuit board. The light emitting diodes **3110** may include at least one of the light emitting diodes according to the exemplary embodiments described above. Light emitted from the light source module enters the light guide plate **3250** and is supplied to the display panel **3210** through the optical sheets **3230**. The light guide plate **3250** and the optical sheets **3230** convert spot light emitted from the light emitting diodes **3110** into sheet light.

In this way, the light emitting diodes according to the exemplary embodiments may be applied to edge type displays like the display apparatus according to the present embodiment.

FIG. **25** is a cross-sectional view illustrating a head light to which a light emitting diode according to another exemplary embodiment of the inventive concepts is applied.

Referring to FIG. **25**, the headlight according to the present embodiment includes a lamp body **4070**, a substrate **4020**, a light emitting diode **4010**, and a cover lens **4050**. The headlight may further include a heat dissipation unit **4030**, a support rack **4060**, and a connection member **4040**.

The substrate **4020** is secured by the support rack **4060** and is disposed above the lamp body **4070**. As the substrate **4020**, any member capable of supporting the light emitting diode **4010** may be used without limitation. For example, the substrate **4020** may include a substrate having a conductive pattern, such as a printed circuit board. The light emitting diode **4010** is disposed on the substrate **4020** and may be supported and secured by the substrate **4020**. In addition, the light emitting diode **4010** may be electrically connected to an external power source through the conductive pattern of the substrate **4020**. Further, the light emitting diode **4010** may include at least one of the light emitting diodes according to the exemplary embodiments described above.

The cover lens **4050** is disposed on a path of light emitted from the light emitting diode **4010**. For example, as shown in the drawing, the cover lens **4050** may be separated from the light emitting diode **4010** by the connection member **4040** and may be disposed in a direction of supplying light emitted from the light emitting diode **4010**. By the cover lens **4050**, an orientation angle and/or a color of light emitted by the headlight can be adjusted. On the other hand, the connection member **4040** is disposed to secure the cover lens **4050** to the substrate **4020** while surrounding the light emitting diode **4010**, and thus can act as a light guide that provides a luminous path **4045**. The connection member **4040** may be formed of a light reflective material or coated therewith. On the other hand, the heat dissipation unit **4030** may include heat dissipation fins **4031** and/or a heat dissipation fan **4033** to dissipate heat generated upon operation of the light emitting diode **4010**.

In this way, the light emitting diodes according to the exemplary embodiments may be applied to headlights like the headlight according to the present embodiment, particularly, vehicular headlights.

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Although the various exemplary embodiments of the inventive concepts have been described above, the inventive concepts are not limited to these embodiments, and various modifications can be made. In addition, the elements described in one embodiment can be applied to other embodiments without departing from the technical spirit according to the appended claims of the inventive concepts.

What is claimed:

1. A light apparatus, comprising:

a substrate including a conductive pattern;  
a light source disposed on the substrate and electrically connected to the conductive pattern, the light source including:

a first conductivity type semiconductor layer;  
a mesa disposed on the first conductivity type semiconductor layer, the mesa comprising an active layer and a second conductivity type semiconductor layer;  
an ohmic layer disposed on the mesa and electrically connected to the second conductivity type semiconductor layer;

a lower insulation layer disposed on the ohmic layer and including a first material and a second material;  
a preliminary layer disposed between the ohmic layer and the lower insulation layer;

a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer;

a second pad metal layer disposed on the lower insulation layer, and electrically connected to the ohmic layer;

an upper insulation layer covering the first pad metal layer and the second pad metal layer, the upper insulation layer comprising a first opening configured to expose the first pad metal layer and a second opening configured to expose the second pad metal layer;

a first bump pad connected to the first pad metal layer through the first opening of the upper insulation layer; and

a second bump pad connected to the second pad metal layer through the second opening of the upper insulation layer; and

a light transmissive layer disposed on the light source, wherein the lower insulation layer has a plurality of refractive indices, and the upper insulation layer covers a side surface of the preliminary layer and the first material of the lower insulation layer.

2. The light apparatus of claim 1, wherein the preliminary layer and upper insulation layer include a material identical to the first material.

3. The light apparatus of claim 1, wherein the mesa has an indent portion intruding therein from one side edge of the mesa.

4. The light apparatus of claim 1, further comprising a substrate, wherein the substrate is rectangular or square.

5. The light apparatus of claim 1, wherein the first material is SiO<sub>2</sub>, the preliminary layer is SiO<sub>2</sub>, and the upper insulation layer is SiO<sub>2</sub>.

6. The light apparatus of claim 1, wherein the first pad metal layer and the second pad metal layer include a multilayer structure having a plurality of metal layers.

7. The light apparatus of claim 1, wherein the ohmic layer includes a transparent layer and a metal layer having reflectance to a light emitted from the active layer.

8. The light apparatus of claim 7, wherein the ohmic layer includes a transparent layer and a metal layer having reflectance to a light emitted from the active layer.

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9. A light apparatus, comprising:  
 a substrate including a conductive pattern;  
 a light source disposed on the substrate and electrically connected to the conductive pattern, the light source including:  
 a first conductivity type semiconductor layer;  
 a mesa disposed on the first conductivity type semiconductor layer, the mesa comprising an active layer and a second conductivity type semiconductor layer;  
 an ohmic layer disposed on the mesa and electrically connected to the second conductivity type semiconductor layer;  
 a lower insulation layer disposed on the ohmic layer;  
 a preliminary layer disposed between the ohmic layer and the lower insulation layer;  
 a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer;  
 a second pad metal layer disposed on the lower insulation layer, and electrically connected to the ohmic layer;  
 an upper insulation layer covering the first pad metal layer and the second pad metal layer, the upper insulation layer comprising a first opening configured to expose the first pad metal layer and a second opening configured to expose the second pad metal layer;  
 a first bump pad connected to the first pad metal layer through the first opening of the upper insulation layer; and  
 a second bump pad connected to the second pad metal layer through the second opening of the upper insulation layer; and  
 a light transmissive layer disposed on the light source, wherein the lower insulation layer includes a first material and a second material, and the preliminary layer and the upper insulation layer include a material identical to the first material.
10. The light apparatus of claim 9, wherein the mesa has an indent portion intruding therein from one side edge of the mesa.
11. The light apparatus of claim 9, further comprising a substrate, wherein the substrate is rectangular.
12. The light apparatus of claim 9, wherein the upper insulation layer covers a side surface of the preliminary layer and the first material of the lower insulation layer.
13. The light apparatus of claim 9, wherein the first material is SiO<sub>2</sub>, the preliminary layer is SiO<sub>2</sub>, and the upper insulation layer is SiO<sub>2</sub>.
14. The light apparatus of claim 9, wherein the first pad metal layer and the second pad metal layer include a multilayer structure having a plurality of metal layers.

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15. A light apparatus, comprising:  
 a substrate including a conductive pattern;  
 a light source disposed on the substrate and electrically connected to the conductive pattern, the light source including:  
 a first conductivity type semiconductor layer;  
 a mesa disposed on the first conductivity type semiconductor layer, the mesa comprising an active layer and a second conductivity type semiconductor layer;  
 an ohmic layer disposed on the mesa and electrically connected to the second conductivity type semiconductor layer;  
 a lower insulation layer disposed on the ohmic layer and including a first material and a second material;  
 a preliminary layer is disposed between the ohmic layer and the lower insulation layer;  
 a first pad metal layer disposed on the lower insulation layer and electrically connected to the first conductivity type semiconductor layer;  
 a second pad metal layer disposed on the lower insulation layer, and electrically connected to the ohmic layer;  
 an upper insulation layer covering the first pad metal layer and the second pad metal layer, the upper insulation layer comprising a first opening configured to expose the first pad metal layer and a second opening configured to expose the second pad metal layer;  
 a first bump pad connected to the first pad metal layer through the first opening of the upper insulation layer; and  
 a second bump pad connected to the second pad metal layer through the second opening of the upper insulation layer; and  
 a light transmissive layer disposed on the light source; wherein the lower insulation layer has a reflectivity and a plurality of refractive indices, and the preliminary layer and the upper insulation layer include a material identical to the first material.
16. The light apparatus of claim 15, wherein the mesa has an indent portion intruding therein from one side edge of the mesa.
17. The light apparatus of claim 15, wherein the upper insulation layer covers a side surface of the preliminary layer and the first material of the lower insulation layer.
18. The light apparatus of claim 15, wherein the first material is SiO<sub>2</sub>, the preliminary layer is SiO<sub>2</sub>, and the upper insulation layer is SiO<sub>2</sub>.
19. The light apparatus of claim 15, wherein the first pad metal layer and the second pad metal layer include a multilayer structure having a plurality of metal layers.
20. The light apparatus of claim 15, wherein the ohmic layer includes a transparent layer and a metal layer having reflectance to a light emitted from the active layer.

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